Mitigating Evaluative Epistemic Filtering in LLM-Driven Essay Scoring: A Culturally-Aware Multi-Perspective Scoring (CAMPS) Framework

by

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Declaration

I certify that except where due acknowledgement has been made, this thesis is my own work, that any help received in preparing this thesis has been acknowledged, and that all sources used have been indicated. I certify that the work is original and has not been submitted previously, in whole or in part, for any other academic award. Ethics procedures and guidelines have been followed (RMIT Human Research Ethics exemption, Project 31036). Any use of generative AI tools in the preparation of this thesis has been disclosed in accordance with course policy. Student name: Bao Nguyen

Student number: s4139514

Signature: ....

Date: ....

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Use of generative AI. In accordance with course policy, the use of generative AI tools in this project is disclosed here. AI assistance (principally Claude, Anthropic) was used for software engineering support in the experiment pipeline and analysis scripts, for copy-editing and restructuring of prose drafted by the author, for checking citations against primary sources, and for examiner-style review of draft versions, whose comments the author evaluated and acted on. The research questions, framework design, experimental decisions, data collection, interpretation of results, and conclusions are the author's own; no experimental result was generated by an AI tool other than the LLM evaluators that are the object of study, and every statistic is regenerated by script from logged model outputs.

Summary

Imagine a Vietnamese student who opens her essay with a proverb, “water wears down stone”, and argues as many Asian traditions teach: context first, conclusion arrived at rather than announced. An AI grader trained mostly on Western text may see only a missing thesis statement and mark her down.

AI now grades high-stakes essays. This thesis measures that gap and tests CAMPS: a panel of AI judges, each reading the essay through a different cultural lens, with their disagreement flagging the essay for a teacher.

Every AI model tested scored native-English essays above upper-level Southeast Asian ones, the widest gap over a full band. Since no human had judged the groups to be of equal quality, this is a disparity, not a proven cultural bias. The panel narrowed the gap by about a quarter on two models, widened it on one, and more judges did not help; its disagreement signal did not pick out the essays needing a human, likely because it is coarse and loosely tied to cultural disagreement. Any AI grader’s gap can now be measured before it is trusted, and neither a single judge nor a diverse panel should be presumed fair without such a test.

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Abstract

Large language models (LLMs) increasingly score essays, yet models trained mainly on Western text may treat Anglo-American rhetoric as the standard and penalise other valid traditions. This thesis names the mechanism *evaluative epistemic filtering* and develops CAMPS (Culturally-Aware Multi-Perspective Scoring), an inference-time framework whose agents, each calibrated to a distinct cultural reading of the rubric, score each essay independently, a Bias Divergence Score (BDS) measuring disagreement. Across 200 topic-controlled ICNALE essays and four model families, every baseline model scored native above upper-band Southeast Asian writers (0.19–1.18 bands; intervals exclude zero). Unmatched on judged quality, the cohorts show an unadjusted disparity, not an isolated cultural bias. A three-agent panel reduced it by 23–28% on the two most disparate models, marking both cohorts down (natives more) and no more than its Southeast Asian lens alone, and amplified it on the least disparate one. Five agents gave no clear improvement on GPT-4o and widened the gap on Claude. Reweighting showed equal weighting inefficient and a small trade-off with band association. At thresholds set for 5% calibration false positives, BDS reached held-out sensitivity of 0.06 and 0.00. Culturally framed prompts change LLM scoring model-dependently; cohort gaps and agent disagreement are imperfect fairness proxies.

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Abbreviations

ASAP	Automated Student Assessment Prize (essay dataset)
AUC	Area Under the ROC Curve
AES	Automated Essay Scoring
AI	Artificial Intelligence
API	Application Programming Interface
BDS	Bias Divergence Score
CAMPS	Culturally-Aware Multi-Perspective Scoring
CEFR	Common European Framework of Reference for Languages
CI	Confidence Interval
CRI	Cultural Rubric Interpretation
ENS	English Native Speaker (ICNALE cohort)
FPR	False Positive Rate
GRA	Global Rating Archives (ICNALE module)
ICNALE	International Corpus Network of Asian Learners of English
IELTS	International English Language Testing System
L1	First (native) language
LLM	Large Language Model
QWK	Quadratic Weighted Kappa
ROC	Receiver Operating Characteristic
SD	Standard Deviation
SEA	Southeast Asia(n)
TOEFL	Test of English as a Foreign Language
WE	Written Essays (ICNALE module)
WEIRD	Western, Educated, Industrialised, Rich, Democratic
WEP	Written Essays Plus (ICNALE module)

Chapter 1

Introduction

1.1 Problem Statement

Automated Essay Scoring (AES) has entered a new era. Where earlier systems counted surface features such as word length, sentence complexity, and spelling, large language models (LLMs) now evaluate what previous systems could not: the quality of a student's argument. Frontier models are increasingly deployed to score high-stakes writing at scale, promising consistent, rapid, and inexpensive assessment. Yet a model that judges *how well a student argues* must hold some standard of what good argumentation looks like, and that standard is not neutral. LLMs derive their evaluative expectations from training corpora that overwhelmingly represent Western, Educated, Industrialised, Rich, and Democratic (WEIRD) cultural contexts; empirical work has shown that widely used models express cultural values closely aligned with English-speaking and Protestant-European countries [Tao et al., 2024].

Consider an illustration. A Vietnamese student opens an argumentative essay with her grandfather's proverb, *nuoc chay da mon* (water wears down stone). She then develops her position the way many Asian academic writing traditions teach: context first, evidence accumulated gradually, the conclusion arrived at rather than announced. A trained human examiner can recognise the coherence of this inductive structure. An LLM evaluator calibrated by its training distribution to expect a thesis statement in the opening paragraph may instead record "lacks a clear thesis" and "argument feels indirect". It may also penalise the hedging forms that signal politeness in her tradition as weak reasoning. If so, she loses marks not because her English is poor, but because the shape of her reasoning is culturally unfamiliar to the model. Figure 1-1 sketches the two shapes.

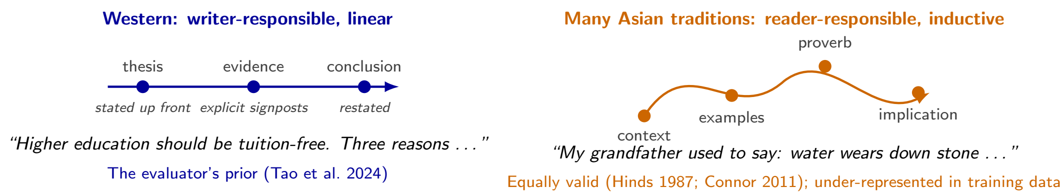

Figure 1-1: Two valid shapes of argument. The writer-responsible, thesis-first pattern is hypothesised to dominate the text LLM evaluators learn from; the corpus composition itself is not measured here, and the measured Western value alignment of these models [Tao et al., 2024] motivates rather than establishes it. The reader-responsible, context-first pattern common in many Asian academic traditions [Hinds, 1987, Connor, 2011] is equally valid but under-represented.

This thesis names this mechanism *evaluative epistemic filtering*: when an LLM evaluates writing, it applies an implicit cultural filter derived from its training distribution, systematically discounting rhetorical strategies that deviate from Anglo-American conventions regardless of their argumentative merit. A pilot study in the preceding phase of this research (Part A) reported scoring disparities between Southeast Asian (SEA) and native-English cohorts on a 30-essay sample across two frontier models, and that a culturally-diversified scoring panel reduced them. The pilot’s corpus did not support a controlled comparison, so Part B rebuilds the data (Section 3.4); its findings motivated, but do not establish, the claims this thesis investigates at scale.

The problem this thesis addresses can therefore be stated plainly. If LLM-based essay scoring applies culturally partial standards, it does so invisibly and at scale, to students whose academic futures depend on the result. Scoring disparities against non-native writers are documented, but no validated framework exists to mitigate them in evaluative settings, and no calibrated mechanism exists to flag the essays affected. This thesis measures the native–SEA scoring disparity across models, builds and tests such a framework, and tests its flagging mechanism against human ground truth. Throughout, *disparity* or *gap* names what is measured (a difference in mean score between cohorts) and *filtering* names the mechanism hypothesised to produce part of it (Table 1.2).

1.2 Research Questions and Hypotheses

Two research questions drive this thesis:

RQ1. To what extent does evaluative epistemic filtering manifest as measurable scoring disparities against Southeast Asian rhetorical patterns in LLM-driven essay scoring, and does it generalise across model families?

RQ2. Can a culturally-aware multi-agent scoring architecture mitigate these disparities at inference time, without retraining or access to model weights, while reliably flagging for human review the essays it cannot score with confidence?

These questions are operationalised as four testable hypotheses:

- **H1 (Generalisation).** A native−SEA cohort score gap, consistent with evaluative epistemic filtering, has a 95% confidence interval excluding zero on at least four LLMs, including an open-weight model.
- **H2 (Mitigation).** A five-agent culturally-calibrated panel with a meta-adjudicator reduces the cohort score gap more than the three-agent pilot configuration, without degrading agreement with the corpus's proficiency bands.
- **H3 (Detection).** A Bias Divergence Score threshold calibrated by receiver operating characteristic (ROC) analysis attains at least 90% sensitivity at no more than a 5% false-positive rate against multi-rater human disagreement on the 140-essay Global Rating Archives (GRA) validation set.
- **H4 (Trade-off boundary).** Consensus-weight allocation exhibits an identifiable boundary beyond which cohort-fairness gains degrade scoring accuracy against graded ground truth.

Graded ground truth does not exist for the cohort essays, so H4 is examined on a proxy, association with the corpus's proficiency bands (Section 3.8), and is reported as exploratory.

The research questions and hypotheses relate as follows. RQ1 is a question about the *existence and generality* of the problem; it is operationalised by H1, which makes it falsifiable by fixing the scope (four model families) and the test (a confidence interval on the cohort scoring gap that excludes zero). RQ2 is a question about the *solution*, and it decomposes into three parts: whether the mitigation works (H2), whether its failures can be detected (H3), and what the mitigation costs (H4). Table 1.1 summarises the mapping and where each hypothesis is tested.

Table 1.1: How the research questions decompose into hypotheses, and where each is tested.

RQ	Hypothesis	Establishes	Tested in
RQ1	H1	A cohort gap exists and generalises across models	E1
RQ2	H2	The panel narrows the gap	E1–E2
RQ2	H3	BDS flags contested essays	E3
RQ2	H4	What narrowing the gap costs (proxy: band association)	E2 (proxy only)

Chapter 5 returns to each hypothesis explicitly. Because several terms recur throughout, Table 1.2 defines them once in plain language.

Table 1.2: Key terms used in this thesis, in plain language.

Term	Meaning in this thesis
Evaluative epistemic filtering	The tendency of an AI grader to mark down writing whose rhetorical form is unfamiliar from its training data, regardless of the argument's quality.
Lens	One AI evaluator given a documented cultural interpretation of the same scoring rubric (Western, Southeast Asian, East Asian, Mekong, or Neutral). "Culturally calibrated" means informed by that scholarship, not empirically tuned.
Panel, consensus	Several lenses scoring the same essay independently; their weighted average is the consensus score.
Bias Divergence Score (BDS)	How much the lenses disagree on one essay; high values are meant to flag the essay for a human.
Cohort, cohort gap	A group of essays by writer background (Southeast Asian learners or native-English writers); the gap is the difference in mean score between the two cohorts. It is an *unadjusted disparity*: the cohorts were not matched on human-judged quality, so the gap may contain both filtering and real quality differences.
Contested essay	In E3, an essay in the top quartile of disagreement among 80 human raters. This is a proxy for the cultural contestation BDS is meant to detect, not the same thing.
Band	The 1–9 score an evaluator assigns (integer, with occasional half-bands); the corpus's own proficiency labels are CEFR bands (B1_2, B2+).

1.3 Theoretical Framing: Epistemic Injustice and Contrastive Rhetoric

Two established bodies of theory ground this thesis. The first is Fricker [2007]'s account of *testimonial injustice*: a speaker is wronged in their capacity as a knower when prejudice in the hearer deflates the credibility of their testimony. Evaluative epistemic filtering can be understood as testimonial injustice automated and scaled: the student's argument is discounted not on its merits but because its form does not match the evaluator's culturally conditioned expectations. Fricker's framework also cautions against a naive remedy: no evaluator, human or artificial, judges from nowhere. This thesis therefore treats a "culturally neutral" evaluator as a regulatory ideal rather than an achievable design goal, and pursues *plurality* of perspectives rather than pretended neutrality.

The second grounding is contrastive rhetoric. Since Kaplan [1966] first observed that rhetorical organisation is culturally learned, research summarised by Connor [2011] has documented systematic differences between writing traditions. These include the distinction between writer-responsible and reader-responsible coherence, and the validity of inductive, context-first argument structures common in Asian academic writing. This literature supplies the scholarly basis for the cultural rubric interpretations at the centre of the proposed framework, distinguishing them from ad-hoc cultural personas.

The mitigation strategy itself extends the formal multi-perspective bias-reduction framework of Dorleon and Shujaa [2026], who proved that, under ideal conditions, generating text from multiple cultural perspectives with bias filtering can eliminate stereotypical content and perspective omission in *generative* tasks. Companion empirical work documented pervasive cultural bias in LLM treatments of Southeast Asian content [Shujaa et al., 2026]. This thesis asks whether those principles transfer from generation to *evaluation*. Independent support for the core mechanism comes from the evaluation literature: panels of diverse LLM judges have been shown to outperform and de-bias single large judges [Verga et al., 2024], though with diversity defined by model provider rather than by culture. The thesis, in short, tests whether the jury effect holds when the axis of diversity is cultural.

Finally, the fairness–accuracy trade-off at the centre of H4 has a direct lineage in fairness-aware machine learning. Dorleon et al. [2022] formulated feature selection as an explicit trade-off between fairness and prediction performance. FAPFID [Dorleon et al., 2023] then showed that fairness for unprivileged groups can be improved without sacrificing overall performance, through balanced representation. H4 asks whether an analogous frontier exists in evaluative scoring, and where it lies.

1.4 Significance

The significance of this work is threefold. First, equity: if LLM-based assessment comes to gate scholarships, university admission, and migration pathways, as its adoption in essay scoring makes plausible, a systematic penalty against non-Western rhetorical traditions would constitute discrimination embedded in infrastructure, invisible to its subjects and deniable by its operators. Measuring the disparity rigorously is a precondition for governing it. Second, method: existing multi-agent AES research improves accuracy without cultural awareness [e.g., Su et al., 2025], while existing debiasing research targets generative outputs rather than evaluative judgements; a validated framework for *evaluative* fairness would fill a documented gap at their intersection. Third, deployability: because the proposed framework operates entirely at inference time through commercial APIs, any institution can apply it without retraining models. This practical property distinguishes it from training-time approaches and matches the constraints of real educational deployments, particularly in the Southeast Asian context where this research is situated.

1.5 Contributions

This thesis makes five contributions:

1. **Conceptual:** it formalises evaluative epistemic filtering as a distinct bias mechanism in LLM-driven scoring, theoretically grounded in epistemic injustice and contrastive rhetoric.

2. **Architectural:** it develops CAMPS (Culturally-Aware Multi-Perspective Scoring),

a culturally-calibrated scoring panel with a divergence-based review trigger, extending multi-perspective bias reduction from generation to evaluation, and tests whether scaling it to five lenses and an adjudicator helps.

3. **Empirical:** it evaluates the framework on 200 topic-controlled essays from the ICNALE corpus [Ishikawa, 2023] across four frontier and open-weight models, establishing the disparity at scale on a corpus suited to the comparison and characterising where the framework changes it.

4. **Methodological:** it introduces ROC calibration of the Bias Divergence Score against multi-rater human disagreement, replacing the pilot's hand-set threshold.

5. **Practical:** it maps the gap against a proficiency-band association proxy under every panel weighting, giving deploying institutions a re-runnable audit and an explicit account of what narrowing the gap did and did not cost on this evidence.

1.6 Thesis Outline

The thesis follows the research questions in order. Chapter 2 reviews the literature that frames RQ1 and RQ2 and locates the gap this thesis occupies. Chapter 3 develops the CAMPS framework and the method: the data, the three experiments (E1 for H1 and part of H2, E2 for H2 and H4, E3 for H3), the statistical protocol, and the alternatives considered. Chapter 4 reports the experiments in that order and discusses what the results mean, why they arose, how they compare with prior work, and where they are uncertain. Chapter 5 answers each hypothesis, states the contributions and limitations, and outlines future work.

Chapter 2

Literature Review

The literature relevant to this thesis was searched between July and September 2026 using Google Scholar, arXiv, the ACL Anthology, and the Springer and IEEE digital libraries. The search combined the terms "automated essay scoring", "multi-agent", "LLM-as-a-judge", "cultural bias", "debiasing", "fairness", "non-native writers", and "contrastive rhetoric". It was supplemented by backward and forward citation chasing from key papers. Every reference cited in this thesis was verified against its primary source. The review is organised thematically: multi-agent and panel-based scoring (Section 2.1), inference-time debiasing (Section 2.2), cultural bias in LLMs (Section 2.3), fairness in AES for non-native writers (Section 2.4), and the contrastive-rhetoric foundations (Section 2.5), before Section 2.6 locates the gap this thesis addresses.

2.1 Multi-Agent and Panel-Based Approaches to Scoring

Two research communities have converged on the same architectural intuition, that several evaluators are better than one, for different reasons.

In automated essay scoring, multi-agent designs pursue *accuracy*. CAFES [Su et al., 2025] orchestrates scorer, feedback-manager, and reflective-reviser agents (average 21% improvement in Quadratic Weighted Kappa, QWK, over direct prompting). Roundtable Essay Scoring [RES; Jang et al., 2025] consolidates trait-based evaluator agents through simulated discussion (up to 34.86% QWK improvement on the ASAP benchmark, zero-shot). Related designs add debate with retrieval [Keramati et al., 2026] and component recognition [Wang et al., 2025]. What unites this family is the axis along which agents differ: *function* or *trait*. None varies the cultural interpretation of the rubric itself, and none evaluates fairness across writer cohorts. Improvements are reported on aggregate agreement. This leaves open the possibility that a more accurate panel is uniformly accurate for some writers and uniformly harsh for others.

In the LLM-evaluation community, panel designs pursue *bias reduction*. Verga et al. [2024] replaced a single large judge model with a Panel of LLM evaluators (PoLL) drawn from three different model families. They found that the panel not only correlated better with human judgement across six datasets but also reduced intra-model self-preference bias, at roughly a seventh of the cost. This is, to date, the clearest empirical demonstration that judge *diversity* counteracts judge *bias*. The diversity axis, however, is model provider, chosen to decorrelate model-specific quirks rather than to represent any substantive difference in evaluative standards. The present thesis can be read as a targeted transplant. It takes the panel-of-judges mechanism whose bias-reducing effect Verga et al. [2024] established, and re-axes the diversity from provider to *cultural rubric interpretation*. Here the bias to be countered is cultural rather than idiosyncratic.

2.2 Inference-Time Debiasing

A second literature attacks bias at inference time, without retraining. BiasFilter [Cheng et al., 2025] periodically evaluates candidate continuations during generation against a fairness reward model and discards low-reward segments, enforcing fairness on *generated text* for both open-source and API-based models. The framework demonstrates that meaningful debiasing is achievable purely at inference time, a design constraint this thesis shares. But its unit of intervention is the output token stream, which has no analogue when the model's output is a single scalar score.

Closest to the present work is Multi-Agent Cultural Debate (MACD) [Tan et al., 2026], which assigns LLM agents explicit cultural personas and orchestrates deliberation. MACD substantially raised the rate of unbiased responses over single-model baselines, and its central finding, that functionally-roled agent frameworks lack the explicit cultural representation cross-cultural fairness needs, independently corroborates the design principle behind CAMPS. The two frameworks nonetheless answer different questions. MACD debiases *what a model says* in open-ended generation. The same paper also introduces Multi-Agent Vote (MAV), an evaluation protocol in which several agents classify a response's cultural tendency, with an explicit no-bias option, and their votes are aggregated; this is the closest architectural precedent for culturally differentiated judging. CAMPS differs in applying such judging to essay scoring, in analysing the resulting scores cohort by cohort, and in treating disagreement between cultural agents as a candidate human-review signal rather than a vote to be aggregated away.

A caution for all multi-agent designs comes from the MALIBU benchmark [Mirza et al., 2025], which shows that LLM-based multi-agent systems can implicitly *reinforce* social biases, and that persona-conditioned judging can over-correct. For this thesis, MALIBU motivates two design decisions: consensus weights are calibrated empirically rather than assumed benign (Hypothesis H4 examines, exploratorily, how the cohort gap moves with proficiency-band association under reweighting), and panel behaviour is validated against multi-rater human ground truth rather than taken on faith.

A separate line of work mitigates bias inside the model rather than around it. Self-debiasing exploits a model's own ability to recognise undesirable properties of its output, suppressing them at decoding time [Schick et al., 2021], and FairSteer intervenes on hidden-layer activations with dynamically applied steering vectors at inference time [Li et al., 2025]. Both are effective in their settings, but both are white-box methods: they require access to output distributions or internal activations that commercial scoring deployments accessed through an API do not expose. CAMPS was deliberately designed to operate at the level of prompts and outputs alone, trading the precision of internal intervention for applicability to any evaluator, including closed models.

2.3 Cultural Bias in Large Language Models

That LLMs carry a Western-centric value orientation is well documented, although whether it extends to *evaluative* judgement is exactly what this thesis tests. Tao et al. [2024] compared five OpenAI model generations against nationally representative values-survey data for 107 countries. They found that model responses consistently resemble those of English-speaking and Protestant-European populations. This is the WEIRD alignment that this thesis proposes as a distal contributing cause of evaluative epistemic filtering; Tao et al. measure value alignment in responses, not the prevalence of rhetorical forms in training data, so the link is hypothesised.

The proximal framework for this thesis comes from the supervisor's research programme. Dorleon and Shujaa [2026] formally defined bias categories for generative outputs (stereotyping, omission, ethnocentrism, simplification) and proposed multi-perspective generation with bias detection, proving that under ideal conditions the method eliminates stereotypical content and perspective omission. The companion study [Shujaa et al., 2026] evaluated seven state-of-the-art LLMs on prompts embedding culturally sensitive assumptions about Vietnam, Myanmar, and Nepal, documenting pervasive stereotyping, linguistic ethnocentrism, and epistemic bias, with substantial variation across models. The same group's work on framing-bias detection [Shujaa and Dorleon, 2025] reflects the detection-oriented strand of this programme that the Bias Divergence Score continues: bias treated not only as something to remove but as something to *measure and flag*.

Two further studies shaped how cultural calibration is implemented in this thesis. SEA-HELM, a community-built evaluation suite for Southeast Asian languages, argues that benchmarks produced by machine translation or synthetic generation without community input lose cultural authenticity [Susanto et al., 2025]. The same reasoning led this thesis to ground its cultural lenses in the contrastive rhetoric literature and in a corpus built in the region, rather than in ad hoc persona descriptions. More cautionary still, Khan et al. [2025] showed that survey-based measures of LLM cultural alignment are unstable across presentation formats and that prompting a model to represent a specific culture is not reliably steerable. For CAMPS this is a design constraint, not a refutation: lens prompts are fixed verbatim (Appendix A), decoding uses temperature 0, and the panel's behaviour is validated empirically against human raters (E3) rather than assumed correct because the prompt requested it.

2.4 Fairness in AES for Non-Native Writers

Concerns about equitable automated scoring of non-native writing predate the LLM era. Vajjala [2018] showed on the TOEFL11 and FCE corpora that the linguistic features most predictive of essay quality differ across L1 populations, evidence that a single scoring model implicitly encodes population-specific assumptions about what good writing looks like. What has changed with LLM-based scoring is the depth at which such assumptions operate: where feature-based systems encoded them in engineered predictors that could be inspected, LLM evaluators encode them in an opaque prior over rhetorical form [cf. Gallegos et al., 2024].

The LLM-era evidence is accumulating quickly. Pack et al. [2024] evaluated four prominent LLMs (PaLM 2, Claude 2, GPT-3.5, GPT-4) as scorers of English-learner placement essays, finding usable reliability for the strongest model but meaningful variation across models and administrations. Evaluative behaviour is therefore model-specific, which motivates testing H1 across at least four model families rather than generalising from one. Most directly relevant, Gayed [2026] fine-tuned an open-weight model (Gemma-3-27B) for essay scoring and evaluated it on the full TOEFL11 corpus of 12,100 essays across eleven L1 backgrounds, explicitly analysing scoring effects by first language. The study documents L1-linked scoring variation on exactly the corpus family and task that concern this thesis. Like the broader AES fairness literature it extends, however, it stops at *measurement*: no mitigation framework is proposed, and the native/non-native design this thesis requires is out of scope because TOEFL11 contains no native-English cohort. Related recent work has begun to probe mechanism as well as outcome. Yang et al. [2025] showed that prompt-based LLM scorers can infer writer demographics, including English-learner status, from essay text alone. They also showed that scoring behaviour shifts once such a status is recognised.

Taken together, this literature establishes the disparity, its model-dependence, and hints at its mechanism, while leaving the mitigation side of the problem essentially untouched.

That asymmetry between diagnosis and treatment is the practical motivation for the framework developed in Chapter 3.

2.5 Contrastive Rhetoric and Cultural Writing Traditions

The claim that rhetorical organisation is culturally learned, and that an evaluator calibrated to one tradition will therefore systematically misread another, has a research lineage of its own. Kaplan [1966] first argued that paragraph organisation reflects culturally specific "thought patterns", contrasting the linear development expected in English academic prose with the patterns he observed in other traditions. The framework was refined substantially in later work: Hinds [1987] distinguished *writer-responsible* languages, in which the burden of clarity falls on the writer through explicit signposting, from *reader-responsible* languages, in which coherence is expected to be inferred by the reader. Under this typology, the "missing thesis statement" an LLM penalises may simply be reader-responsible coherence operating as designed. Connor [2011] consolidated this field as intercultural rhetoric, correcting early essentialist readings: rhetorical traditions are conventions distributed across writing communities, not fixed properties of individuals. This thesis adopts that distinction explicitly, treating L1 cohorts as proxies for rhetorical tradition rather than for identity.

For Southeast Asian learner writing specifically, the empirical base is anchored by the ICNALE project [Ishikawa, 2023], whose topic-controlled design allows rhetorical variation to be observed under matched task conditions across ten Asian countries and a native-English reference cohort. This literature performs two duties in the present thesis. First, it supplies the content of the cultural rubric interpretations in Chapter 3, where each Cultural Rubric Interpretation (CRI) prompt is grounded in documented rhetorical conventions rather than improvised cultural personas. Second, it supplies the defence against the objection that culturally-calibrated agents merely role-play stereotypes. The calibrations are drawn from a half-century of contrastive-rhetoric scholarship, and their behaviour is tested empirically in Chapter 4.

2.6 The Research Gap

Four literatures converge on the problem this thesis addresses, and each stops short of it in a characteristic way. The multi-agent scoring literature (Section 2.1) established that panels of evaluators outperform single judges (CAFES and RES on accuracy, PoLL on bias). But it defines panel diversity by function, trait, or model provider, never by cultural interpretation of the rubric, and it does not measure fairness across writer cohorts. The inference-time debiasing literature (Section 2.2) established that bias can be mitigated without retraining, and MACD specifically established that *explicit cultural representation* in agent frameworks matters. But its object is generated text, not evaluative judgement, and it dissolves inter-agent disagreement rather than exploiting it. The cultural-bias literature (Section 2.3) established the WEIRD disposition of current models and, in Dorleon and Shujaa [2026], provided a formal multi-perspective mitigation framework with provable guarantees, for generation only. Meanwhile, the AES fairness literature (Section 2.4) continues to document L1-linked scoring disparities without offering a validated mitigation framework.

Within the scope of the searches described above, no published work was found at the intersection: culturally-diverse rubric interpretation, applied to evaluative scoring, with inter-agent divergence used as a calibrated trigger for human review and tested against multi-rater human ground truth. That intersection is where CAMPS operates.

Chapter summary

Two literatures converge on the idea that several evaluators are better than one, but neither varies the *cultural* reading of the rubric, and the fairness literature documents L1-linked scoring disparities without a validated remedy. Chapter 3 builds the framework that fills this gap and the experiments that test it.

Chapter 3

Methodology

This chapter sets out what was done and why. It states the working theory (Section 3.1) and specifies the CAMPS framework (Sections 3.2 and 3.3). It then describes the data (Section 3.4) and the three experiments (Section 3.5), records the alternatives considered and the ethical constraints (Sections 3.6 and 3.7), and sets out the final analytical procedures, distinguishing prespecified decisions from revisions made after the first results were examined (Section 3.8).

3.1 Working Theory: Evaluative Epistemic Filtering

The framework developed in this chapter rests on a working theory of why LLM evaluators systematically under-score culturally unfamiliar writing. This section states that theory precisely, identifies the conditions that sustain the phenomenon, and derives the observable signatures that the experiments in Chapter 4 test.

Definition 3.1 (Evaluative epistemic filtering)**.** *Let E be an essay, $f_{\text{human}}(E)$ the score assigned by a trained human examiner under a rubric R, and $f_{\text{LLM}}(E \mid \theta)$ the score assigned by an LLM evaluator whose parameters θ were estimated from a training distribution $\mathcal{D}$, and let $c(E)$ denote the rhetorical tradition of the essay. Evaluative epistemic filtering occurs when the evaluator's deviation from human judgement differs* between *traditions,*

$$\mathbb{E}[\, f_{\text{LLM}} - f_{\text{human}} \,|\, c(E) = c_1] \;\neq\; \mathbb{E}[\, f_{\text{LLM}} - f_{\text{human}} \,|\, c(E) = c_2]\,, \tag{3.1}$$

with essays whose rhetorical form is under-represented in $\mathcal{D}$ receiving the more deflated scores relative to human judgement, independent of argumentative merit. The contrast is meant at comparable independently assessed quality: conditioning also on $f_{\mathrm{human}}(E) = h$ and comparing traditions at each h excludes a uniform scale compression, which would otherwise produce unequal residuals from unequal quality distributions. The between-tradition contrast is essential: an evaluator that scores every tradition one band too low is miscalibrated but does not filter.

The definition makes two commitments. First, the filter is a property of the *evaluator*, not of the essay: the same essay receives different treatment under evaluators with different training distributions. Second, the deviation is conditional on rhetorical tradition rather than on language proficiency. This is what distinguishes evaluative epistemic filtering from the legitimate scoring of weaker English. It is what the band-restricted cohort design of Section 3.4 is built to approach, and what a supplementary, exploratory human-anchored check (Section 4.4) examines where human ratings exist.

Four conditions are hypothesised to sustain the filter: training corpora over-represent WEIRD contexts, whose value orientation models absorb [Tao et al., 2024]; evaluators' standards are opaque, distributed across parameters [Vajjala, 2018, Gallegos et al., 2024]; diagnostic effort has concentrated on generative outputs [Dorleon and Shujaa, 2026]; and L1-linked scoring disparity is documented [Gayed, 2026] but unconnected to any mitigation mechanism.

The theory yields three testable signatures, mapping onto the hypotheses of Chapter 1. The first is a systematic scoring gap between an upper-band learner cohort and the native reference, consistent with filtering though not separable by this design from residual quality difference (H1). The second is shrinkage of that gap under a panel whose members embody different rubric interpretations, on the premise that their individual filters differ (H2); independent calls do not by themselves make the lenses' errors statistically independent, since they share a base model. The third is measurable panel disagreement on precisely the essays where filtering operates, making divergence itself a detection signal (H3). The fairness–accuracy boundary (H4) follows from the consensus mechanism below.

3.2 The CAMPS Framework

CAMPS (Culturally-Aware Multi-Perspective Scoring) operationalises a single design principle: if no evaluator judges from nowhere (Section 1.3), fairness is pursued through a *plurality* of explicitly declared perspectives rather than a pretended neutrality. In practice this means five cultural rubric interpretations (Table 3.2), each declared in an inspectable system prompt. The framework operates entirely at inference time, needs no retraining or weight access, and treats each LLM as a black box behind an API. Figure 3-1 shows the pipeline; the five components are specified below.

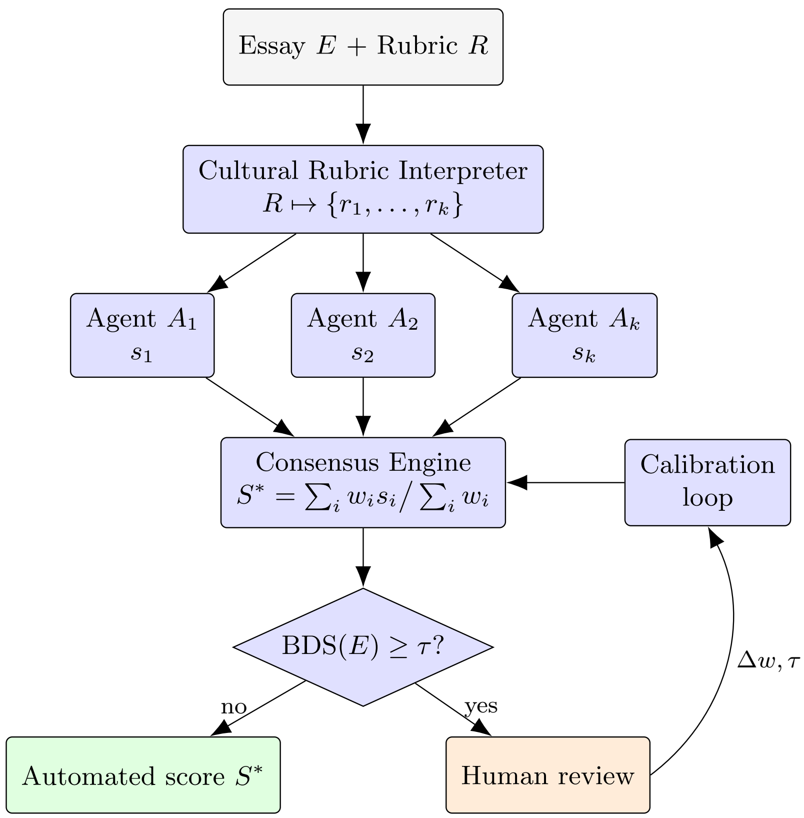

Figure 3-1: The CAMPS pipeline. Each essay is scored in parallel by k *culturally-calibrated* agents: LLM evaluators whose system prompts embed a documented cultural interpretation of the scoring rubric (Component 1). The consensus engine aggregates the scores. The Bias Divergence Score routes contested essays to human review, whose outcomes feed the calibration loop. CAMPS components are shaded blue; inputs and outcomes are unshaded.

Component 1: Cultural Rubric Interpreter (CRI)

Given the scoring rubric R, the CRI produces k culturally-adapted interpretations $\{r_1, \ldots, r_k\}$ through system-level prompting. Each interpretation is grounded in the contrastive-rhetoric literature reviewed in Section 2.5 rather than in improvised personas. The Southeast Asian interpretation, for example, instructs the agent to recognise inductive thesis development and reader-responsible coherence [Hinds, 1987] as valid realisations of the rubric's coherence criterion. The neutral interpretation is retained as a regulatory ideal in Fricker's sense [Fricker, 2007], not as an achievable standpoint. The full prompt texts are reproduced verbatim in Appendix A.

Component 2: Evaluator agent pool

Each interpretation is bound to an independent evaluator agent, and each agent scores the essay without access to the other agents' outputs:

$$s_i = A_i(E, r_i), \qquad i = 1, \ldots, k, \tag{3.2}$$

with temperature $t = 0.0$ and structured (JSON) output; every response is archived, so the analysis is exactly reproducible from the archive even though providers do not guarantee bit-identical regeneration. Independence here is deliberate: MALIBU [Mirza et al., 2025] cautions that interacting agents can amplify rather than cancel bias, so CAMPS confines interaction to the aggregation stage.

Component 3: Consensus aggregation

The panel's scores are combined as a weighted mean,

$$S^*(E) = \frac{\sum_{i=1}^{k} w_i \, s_i(E)}{\sum_{i=1}^{k} w_i}, \qquad w_i \geq 0,\ \sum_i w_i > 0, \tag{3.3}$$

with weights initialised equal. Because S^* is linear in the agents' scores, the consensus cohort gap is the same weighted combination of the individual lenses' cohort gaps. Two consequences follow. Reweighting can trade the gap against agreement with graded ground truth, which is the boundary hypothesised in H4. But when every lens's gap has the same sign, the weighting that minimises the absolute gap is a single lens, the one with the smallest gap (any tie admits mixed optima); a minimum-gap result therefore reflects the aggregation rule, not evidence that combining perspectives improves fairness. Section 3.5 describes how the weight space is swept, and Section 4.2 reports the individual lenses alongside the panel for exactly this reason.

Component 4: Bias Divergence Score (BDS)

The framework's detection signal is the dispersion of the panel:

$$\mathrm{BDS}(E) = \sqrt{\frac{1}{k}\sum_{i=1}^{k}\bigl(s_i(E) - S^*(E)\bigr)^2}. \tag{3.4}$$

BDS is the dispersion of the prompt-conditioned scores around the consensus (computed with equal weights in all experiments). A low BDS indicates agreement among the lenses; agreement alone does not establish that the score is accurate or fair, since all lenses may share a bias. A high BDS means the lenses disagree, which the working theory proposes as the signature of an essay whose score depends on culturally contingent expectations; whether that interpretation holds is what E3 tests. Under the framework, such an essay is flagged for human review. Part B replaces the pilot's hand-set threshold ($\tau = 0.75$) with one calibrated by ROC analysis against multi-rater human ground truth (H3; Section 3.5).

Component 5: Calibration loop

Outcomes of human review feed back into the framework in two ways: the consensus weights w_i are re-optimised to maximise agreement with human scores subject to a cohort-fairness constraint, and the threshold τ is re-estimated as review outcomes accumulate. The loop closes the system without touching model weights; everything learned lives in w and τ. The loop is specified as part of the framework but not exercised in this thesis, whose experiments evaluate a single calibration pass (E2's weight sweep and E3's threshold calibration).

The five components divide the framework's labour: the first two create and apply the plurality of perspectives, the third converts plurality into a single defensible score, and the last two turn residual disagreement into detection and adaptation. The design intends the panel's fairness properties to be a property of the assembly rather than of any one lens; whether they are is tested directly in Section 4.2. Table 3.1 positions CAMPS against the frameworks reviewed in Chapter 2 along the five capabilities the design combines.

Table 3.1: Architectural features compared (what each framework is designed to do, not what has been validated). *Inf.* = inference-time only; *Cult.* = explicit cultural perspectives; *Eval.* = evaluative (scoring) task; *Div.* = divergence-based detection; *HITL* = a threshold-calibrated human-in-the-loop trigger. For CAMPS the detection and trigger features are implemented and evaluated in E3; their operational validity is a finding of Chapter 4, not a premise.

Framework	Inf.	Cult.	Eval.	Div.	HITL
CultureLLM [Li et al., 2024]	×	✓	×	×	×
BiasFilter [Cheng et al., 2025]	✓	×	×	×	×
CAFES [Su et al., 2025]	✓	×	✓	×	×
RES [Jang et al., 2025]	✓	×	✓	×	×
PoLL [Verga et al., 2024]	✓	×	✓	×	×
MACD [Tan et al., 2026]	✓	✓	×	×	×
Dorleon & Shujaa [2026]	✓	✓	×	×	×
CAMPS (this thesis)	✓	✓	✓	✓	✓

Relative to the Part A pilot, the framework specified here differs in three respects, each developed in the following sections. The panel is scaled from three to five cultural interpretations with a meta-adjudicator (Section 3.3). The corpus and cohort design are rebuilt on data that supports the comparison (Section 3.4). Both the consensus weights and the BDS threshold are calibrated empirically rather than hand-set (Section 3.5).

3.3 Scaling the Panel: from Three to Five Agents

The pilot panel comprised three rubric interpretations: Western, Southeast Asian, and a neutral regulatory ideal. Scaling to five requires a principled answer to *which* lenses to add. A candidate lens must satisfy three criteria. It must be (i) *documented* in the contrastive-rhetoric literature, so the CRI prompt encodes scholarship rather than stereotype; (ii) *validatable* against a corpus cohort from the corresponding tradition; and (iii) *distinct*, expected to diverge from the existing panel on identifiable features. These criteria select an *East Asian* lens, grounded in reader-responsible coherence [Hinds, 1987] and validatable against ICNALE's East Asian cohorts, and a *Mekong/Vietnamese* lens, motivated by the region this research serves and validatable against the WEP cohorts. A Middle Eastern lens fails the validatable criterion for this corpus and is deferred. The Mekong lens is a partial exception to criterion (i): the contrastive-rhetoric literature on Mekong-region academic writing is thin, and the lens is admitted on regional relevance (Table 3.2). Separate validation of the two added lenses was outside the scope of E1–E3. Table 3.2 summarises the panel.

Table 3.2: The five-lens panel: rhetorical grounding and the corpus cohort against which each lens's behaviour can be validated.

Lens	Rhetorical grounding	Validating cohort (available)
$\text{CRI}_{\text{Western}}$	Writer-responsible coherence; explicit thesis-first linear development [Kaplan, 1966, Connor, 2011]	ICNALE ENS
CRI_{SEA}	Inductive, context-first development; collectivist framing [Connor, 2011, Ishikawa, 2023]	ICNALE THA, IDN, PHL
$\text{CRI}_{\text{EastAsian}}$	Reader-responsible coherence; delayed thesis; *ki-sho-ten-ketsu*-style organisation [Hinds, 1987]	ICNALE CHN, JPN, KOR, TWN
$\text{CRI}_{\text{Mekong}}$	Proverb- and narrative-anchored argumentation; scholarly motivation from the ICNALE Guide and the supervisor's cultural-bias work [Ishikawa, 2023, Shujaa et al., 2026], not an empirical study of Mekong learner rhetoric	ICNALE WEP VNM, KHM, LAO, MMR
$\text{CRI}_{\text{Neutral}}$	Regulatory ideal: argumentative merit irrespective of structural ordering [Fricker, 2007]	(all cohorts)

The second architectural addition is a *meta-adjudicator*: a further LLM instance that receives the essay, the five agents' scores and their short written rationales, and produces a reasoned consensus score,

$$S^{\text{adj}}(E) = M\big(E, s_1, \rho_1, \ldots, s_k, \rho_k\big), \tag{3.5}$$

where ρ_i is agent i's rationale. The adjudicator is blind to writer identity and cohort metadata and runs at $t = 0.0$. Where the weighted mean treats every judgement as exchangeable evidence, the adjudicator can weigh *arguments*, discounting an agent whose rationale misreads the essay. E2 compares the two aggregation paths directly, with BDS computed from raw agent scores in both conditions; following Mirza et al. [2025], deliberation comes *after* independent judgement, never during scoring.

3.4 Data

Corpus selection

The design requires a corpus with four properties: Southeast Asian learner essays identified by L1 or country; a native-English reference cohort writing under identical task conditions; a human-assigned proficiency measure; and licence terms permitting research at this scale. A systematic search of the major public learner corpora found exactly one satisfying all four. TOEFL11 [Blanchard et al., 2013] has no Southeast Asian L1 and no native cohort, ICLE v3 lacks Southeast Asian L1s, ELLIPSE lacks L1 labels, and EFCAMDAT lacks a native reference.[1] The ICNALE corpus family [Ishikawa, 2023] satisfies all four and is adopted.

The ICNALE corpus family

ICNALE collects topic-controlled English essays from college students across ten Asian countries and regions, together with a native-English reference cohort writing under identical conditions; the three modules used are summarised in Table 3.3. Critically, every

[1] Corpus contents verified against their official distribution pages in July 2026: TOEFL11 via the Linguistic Data Consortium (LDC2014T06), ICLE via UCLouvain, ELLIPSE via its public repository, EFCAMDAT via the EF–Cambridge Open Language Database.

written essay addresses the same two prompts under controlled length and time conditions, removing the prompt-variation confound of TOEFL11's eight-topic design, and all learners carry CEFR-linked proficiency bands (A2–B2+), enabling band-restricted learner-cohort construction.

Table 3.3: ICNALE modules used in this thesis (contents as published on the official ICNALE distribution site, March 2026).

Module	Contents	Role in this thesis
Written Essays (WE) v2.6	5,600 essays, 200–300 words, two controlled topics; ten Asian countries/regions plus native ENS cohort	Primary cohorts: SEA learners and native reference
Written Essays Plus (WEP) v0.7	2,340 essays from additional countries including Vietnam, Cambodia, Laos, Myanmar	Available validating cohort for $\mathrm{CRI}_{\mathrm{Mekong}}$; not used in E1–E3
Global Rating Archives (GRA) v2.1	140 essays, each rated by 80 raters of varied L1 and occupational backgrounds (11,200 ratings)	Multi-rater human ground truth for BDS validation (E3)

Cohort construction and sampling

Figure 3-2 summarises the sample. The Southeast Asian cohort draws from the Thailand, Indonesia, and Philippines components of WE, restricted to the two upper proficiency bands (B1_2, B2+) and stratified by topic and band. The native cohort draws from ENS, which ICNALE does not band (native writers are the corpus's reference group). ICNALE's native module also contains teachers and other adults, but every sampled native essay is by a writer the corpus classifies as a student (USA 83, Canada 8, New Zealand 4, UK 3, Australia 2; ages 19–29), so both cohorts are university-age students. Sampling yields $n = 100$ per cohort (50 per topic) under a fixed seed. Restricting the learner cohort to its upper bands reduces how much of a cohort gap can be attributed to weak English, but it cannot equate the cohorts: any residual quality difference between upper-band learners and native writers remains inside the gap, which is therefore an unadjusted disparity (Section 3.8). A supplementary human-anchored check on the GRA essays, where human ratings exist, is reported in Section 4.4. The GRA's 140 essays are retained in full as E3's validation set; 13 of them (10 SEA, 3 native) also fall in the E1 sample.

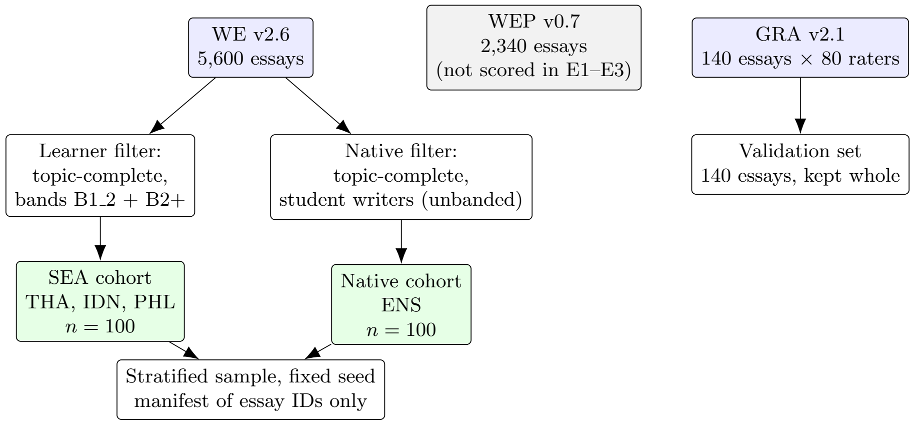

Figure 3-2: Cohort construction and sampling. Essay texts remain within licensed storage; the reproducibility manifest records essay identifiers only.

Scoring target and ground truth

Every agent, including the single-agent baseline, is asked for a band from 1 to 9 on the IELTS Coherence and Cohesion descriptors (the prompts are verbatim in Appendix A). The descriptors are named, not reproduced, in the prompts, so scores rest partly on each model's learned reading of the rubric and are not examiner-equivalent IELTS bands. Claude Sonnet and Gemini Pro returned whole bands only, while Qwen3 returned half-bands (e.g. 6.5) in 295 of its 800 scoring records and GPT-4o in 34 of its 3,700. Returned values were retained as returned, so the score resolution is one band with occasional half-bands. Panel consensus is the mean of the lens scores and so takes fractional values. The corpus's CEFR bands are a proficiency placement (test or vocabulary-size based), not a human rating of the particular essay; association with them is therefore reported as a *proficiency-band association*, not as essay-scoring accuracy. Human ratings of the specific essays exist only for the GRA validation set. Thirteen of the 200 cohort essays also appear in the GRA set; the two analyses are reported separately. In the GRA archive each rater scores ten analytic dimensions (0–10) grouped as Language, Content, and Attitude, plus a holistic score out of 100. Per-essay rating dispersion across the 80 raters provides the human-disagreement benchmark against which BDS is compared. Two properties of the archive shape the E3 design. First, all 140 rated essays address the part-time-job prompt, so the validation set is single-topic. Second, the raters disagree substantially among themselves (mean within-essay standard deviation of the Content ratings ≈ 1.8 on the 0–10 scale). Disagreement among evaluators, human or artificial, is the normal condition of essay assessment. The question E3 asks is whether the panel's divergence tracks the essays on which *humans* diverge.

Licence and handling

ICNALE is distributed free for research with redistribution prohibited [Ishikawa, 2023]. Essay texts are stored only in access-controlled storage and never committed to version control; the reproducibility package releases code, prompts, configurations, and essay-ID manifests only. The data are de-identified by the distributor and fall under the ethics exemption of Section 3.7.

3.5 Experimental Design

Three experiments test the four hypotheses (Table 3.4). E1 runs on all four models; E2 and E3 run on the two models with the largest E1 gaps (GPT-4o and Claude Sonnet), confining the costlier extended conditions to the evaluators where mitigation is material. All conditions share the same settings: temperature fixed at 0.0, structured JSON outputs, model identifiers pinned in the experiment configuration, and a fixed random seed for all sampling. Every result in Chapter 4 is reproducible from the archived model outputs and the released configuration, prompts, and essay-identifier manifest; regenerating the outputs through provider APIs is best-effort (Appendix B).

Experiment E1 (scaled replication) scores both cohorts under the single-agent baseline and the pilot's three-agent panel on all four models, establishing whether the gap exists at scale (H1) and providing the $k{=}3$ reference for E2.

Experiment E2 (extended architecture) scores the same essays under the five-agent panel, with and without the meta-adjudicator. It also sweeps the consensus weights in steps of 0.05 under $\sum_i w_i = 1$, recording the cohort gap and Spearman ρ against CEFR bands at every vector; the full non-dominated set, not a single optimum, is reported and H4's

Table 3.4: Overview of the three experiments: data, conditions, and the hypotheses each tests.

Exp.	Data	Conditions	Tests
E1	SEA + native cohorts ($n = 100$ each)	Single-agent baseline vs. CAMPS k=3, four models	H1; H2 (part)
E2	Same cohorts	CAMPS k=5 with and without meta-adjudicator; consensus-weight sweep ($\Delta w = 0.05$); ablation over $k \in \{1, 3, 5\}$	H2; H4
E3	GRA validation set (140 essays $\times$ 80 raters)	CAMPS k=5; ROC calibration of τ on a calibration split, evaluation on a hold-out split	H3

boundary sought on it. The ablation over k separates the contribution of panel size from that of the adjudicator, and the individual lenses are compared with the panel so that what the panel adds beyond its best single lens is explicit.

Experiment E3 (detection validation) computes BDS for the 140 GRA essays. Human rating dispersion provides the reference notion of a contested essay, operationalised as the top quartile of per-essay standard deviation across the 80 raters' Content ratings, where each rater's Content score is the mean of the four Content dimensions and dispersion is the standard deviation of those per-rater scores across the 80 raters (three essays had 79 usable raters after a missing Content rating was excluded). This is a proxy: GRA raters disagree for many reasons, of which cultural contestation is one, so a detector of cultural divergence is benchmarked here against general disagreement. The 90%/5% operating point was fixed before E3 somewhat below the 96.7% sensitivity the pilot's hand-set threshold had reported, with 5% a conventional false-alarm tolerance. Two thresholds are calibrated on a random half of the GRA set and evaluated on the held-out half: Youden's J (the original analysis), and the constrained rule that H3 actually specifies (the most sensitive threshold whose calibration false-positive rate is at most 5%), added in revision (Section 3.8). Sensitivity and false-positive rate with their counts, and the area under the ROC curve, are reported.

3.6 Alternatives Considered

Training-time cultural adaptation, as in CultureLLM [Li et al., 2024], is excluded on scope and portability grounds: this study evaluates a training-free intervention that can be applied across the selected model interfaces without maintaining separately adapted models for each. *Activation steering* [Li et al., 2025] is infeasible for API-only access (Section 2.2). *Single-prompt debiasing* ("score fairly across cultures") collapses the plurality of standards into one opaque judgement and leaves no divergence signal; it was not separately run. *Inter-agent debate*, as in MACD [Tan et al., 2026], is rejected because interaction can amplify bias [Mirza et al., 2025] and deliberation before scoring destroys the independence BDS needs; CAMPS confines deliberation to the post-hoc adjudicator. *Recruiting human raters* was rejected on ethics-scope grounds (Section 3.7); the GRA supplies eighty raters per essay.

3.7 Ethical Considerations

This research analyses de-identified secondary data only, under an RMIT human research ethics exemption (Project ID 31036). No writer is re-identified, no essay text is redistributed, and no participants are recruited; any extension involving participants would require a new ethics assessment. Two considerations shape the design. First, cultural essentialism: agents are calibrated to *documented rhetorical traditions*, not nationalities, and L1 cohorts are treated as proxies for rhetorical tradition rather than identity. Second, downstream welfare: CAMPS is human-in-the-loop because the cost of a misclassified essay falls on a student. Generative AI use is disclosed in the Acknowledgements.

3.8 Metrics and Statistical Protocol

Primary metric. The primary fairness quantity is the *cohort score gap*: the difference in mean panel score between the native and Southeast Asian cohorts,

$$\Delta = \overline{S^*}_{\text{native}} - \overline{S^*}_{\text{SEA}}, \tag{3.6}$$

computed on an upper-band learner cohort and the unbanded native reference. Δ is an *unadjusted scoring disparity*. On a common scale it decomposes as the human-reference cohort gap plus the difference in model error between cohorts; because the cohorts were not matched on independently assessed essay quality, Δ may include genuine quality differences as well as differential model error, and it does not by itself isolate the cultural mechanism of Definition 3.1. The treatment effects of interest are *changes* in Δ between conditions on the same essays; a fixed quality difference between cohorts enters these only through changes in how a condition maps quality to score (for example, scale compression, visible in the per-cohort standard deviations of Table 4.1). Association with the learner cohort's proficiency bands is reported secondarily as Spearman's ρ between the panel score and the band ordinal on the Southeast Asian cohort. A supplementary human-anchored check on the GRA set is reported in Section 4.4.

Analysis revisions. Three analyses were revised after the first results were examined, and two exploratory analyses added. The proficiency-band analysis originally used quadratic weighted kappa, replaced by Spearman's ρ because the 1–9 score scale and the two-level band labels share no common category system. Cohort uncertainty estimates originally resampled essays, recomputed to resample writers. E3 originally calibrated only Youden's threshold; the constrained rule H3 specifies was added on the same prespecified split after the Youden results had been seen, as were the human-anchored check and the three-lens BDS check on the GRA set. The 90%/5% target, the E3 split, the paired-bootstrap statistic and direction, and the seed were fixed before scoring.

Uncertainty. Every statistic carries its n. ICNALE writers contribute up to two essays (one per topic): the 200 cohort essays come from 161 writers, 39 of whom contribute both, so the writer is the unit of independence, and every cohort gap, gap change and band-association change carries a 95% percentile bootstrap interval that resamples *writers* within each cohort (10,000 resamples, fixed seed), keeping each essay's paired conditions together. Detection statistics carry counts and exact binomial intervals. Intervals are the primary inferential device; the p reported for gap changes is the proportion of bootstrap replicates with reduction ≤ 0, a tail probability, not a null-hypothesis test; nothing was formally registered.

Chapter 4

Results and Discussion

This chapter reports the three experiments in the order of the hypotheses they test, then examines uncertainty and threats to validity, and closes with a discussion of what the results mean, why they arose, and how they compare with prior work. Every statistic is generated by script from the logged runs; every panel-level cohort gap and gap change carries its n and a 95% writer-cluster bootstrap confidence interval (161 writers behind the 200 essays; single-lens gaps are reported as point estimates), band-association changes carry the same interval, and detection statistics are reported with counts and exact binomial intervals. Throughout, the cohort gap is an *unadjusted scoring disparity*: it may include genuine quality differences between the cohorts as well as differential model error (Section 3.8).

4.1 Experiment 1: The Cohort Gap at Scale (H1; H2 in part)

Experiment E1 scored all 200 cohort essays under the single-agent baseline and the three-agent panel on four models; the two conditions together comprised 800 scoring calls per model, all at temperature 0 and logged. Table 4.1 reports the mean and spread of scores per cohort under each condition, the native−SEA score gap with bootstrap confidence intervals, and the gap in standard-deviation units.

Three findings emerge. First, a positive cohort gap exists on every model tested. Under the baseline evaluator, native-cohort essays received higher mean scores than upper-band Southeast Asian essays on all four models, and each gap's 95% confidence interval excludes zero. The gaps were 1.18 bands [0.83, 1.54] for Claude Sonnet, 0.80 [0.59, 1.02] for GPT-4o, 0.23 [0.06, 0.40] for Qwen3 14B, and 0.19 [0.06, 0.31] for Gemini Pro ($n = 100$ per cohort throughout). H1 is supported in the form in which it was tested: an unadjusted native$-$SEA scoring disparity, present on all four model families including an open-weight model. Because the cohorts were not matched on independently assessed essay quality, these gaps do not by themselves isolate the cultural mechanism of Definition 3.1; Section 4.4 examines it, exploratorily, where human ratings exist, between the learner traditions the GRA contains.

Second, the *magnitude* of the gap varied across models, spanning a factor of six in raw bands. Claude Sonnet and GPT-4o exhibited the largest gaps. Gemini Pro and the open-weight Qwen3 showed gaps three to six times smaller, both scoring within a compressed range (Gemini's native scores had a standard deviation of 0.20 bands). In pooled-SD units the spread is narrower, about 2.8-fold (d = 1.09 GPT-4o, 1.01 Claude, 0.43 Gemini, 0.39 Qwen3), so part of the raw heterogeneity is a difference in how much of the scale each model uses. This ordering is only partly consistent with the working theory's claim that the filter reflects the training distribution rather than model quality, since the smallest gap belongs to a Western-trained model. Provenance is also confounded with model scale and score range in this sample, so the ordering is reported as an observation rather than a causal claim.

Third, the panel's effect was model-dependent. Using a writer-cluster paired bootstrap (10,000 resamples), the three-agent panel reduced the gap by 0.27 bands [0.07, 0.49], or 23% ($p = 0.005$), on Claude Sonnet, and by 0.23 bands [0.12, 0.33], or 28% (from unrounded values; $p < 0.0001$), on GPT-4o. On Qwen3 14B the change was -0.04 bands [-0.16, 0.08], no clear evidence of an effect. On Gemini Pro the panel *amplified* the gap, from 0.19 to 0.45 bands (change -0.26 [-0.39, -0.13]). The lens-level scores locate the mechanism. Within Gemini's panel the neutral lens scored far higher than the baseline evaluator and with a much larger cohort gap (7.18 native vs. 6.33 SEA, a 0.85-band gap on its own; $n = 100$ per cohort), while the Western and Southeast Asian lenses remained compressed. On this model the lens prompts shifted how much of the score range was used, not merely which rhetorical features were rewarded, and the panel mean inherited the expanded lens's gap. This is the behaviour the MALIBU benchmark warns of for persona-conditioned judging. Gemini was not carried into E2, so it bears on H2 only as a boundary condition: the panel is not a safe default on a model whose baseline gap is already small.

Figure 4-1 visualises the primary result: the gap on every model under both conditions, with its confidence intervals.

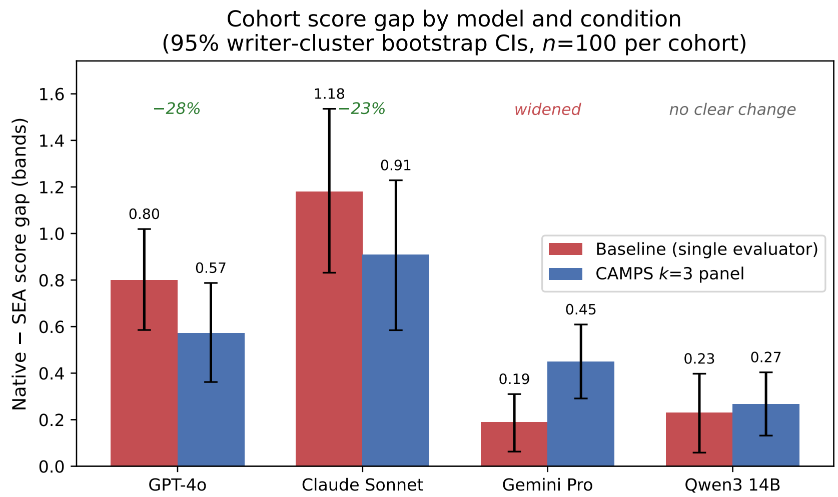

Figure 4-1: Native−SEA score gap by model and condition, with 95% writer-cluster bootstrap confidence intervals (n=100 per cohort). Every baseline gap excludes zero; the panel narrows the gap on the two models where it is largest, shows no clear change on Qwen3, and amplifies it on Gemini Pro.

The pattern in these results anticipates the fuller treatment in Sections 4.2 and 4.6: the panel's effect differed by model, with the two largest reductions on the two models with the largest baseline gaps, no clear effect on Qwen3, and amplification on Gemini Pro. Cultural plurality in the evaluator is therefore not an intervention to be applied without a baseline audit of the kind E1 provides. Figure 4-2 summarises the four model-level observations; because the baseline gap enters both axes, and only four models were tested, the figure is descriptive and no dose–response mechanism is inferred from it.

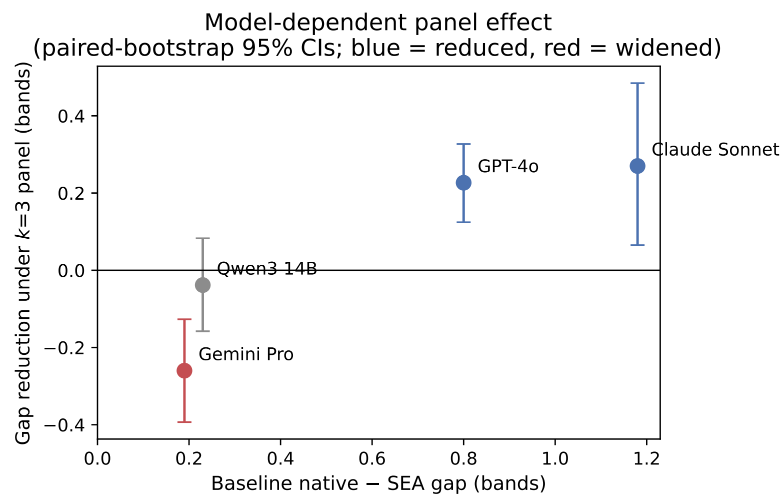

Figure 4-2: Gap reduction under the three-lens panel against the baseline gap, one point per model, with writer-cluster paired-bootstrap 95% confidence intervals. The panel reduced the gap on the two models with the largest baseline gaps, had no clear effect on Qwen3, and amplified the gap on Gemini Pro. Descriptive: four observations, with the baseline gap on both axes.

Table 4.1: Experiment E1: mean consensus score (SD) by cohort, $n = 100$ per cohort, and the native−SEA score gap (upper-band SEA cohort vs unbanded native reference) with 95% writer-cluster bootstrap confidence intervals (10,000 resamples); d is the gap in pooled-SD units.

Model	Condition	Mean_{SEA} (SD)	$\text{Mean}_{\text{native}}$ (SD)	Gap [95% CI]	d
Claude Sonnet	Baseline	4.44 (1.39)	5.62 (0.91)	1.18 [0.83, 1.54]	1.01
Claude Sonnet	CAMPS k=3	3.73 (1.14)	4.64 (1.04)	0.91 [0.58, 1.23]	0.83
Gemini Pro	Baseline	4.83 (0.59)	5.02 (0.20)	0.19 [0.06, 0.31]	0.43
Gemini Pro	CAMPS k=3	5.24 (0.62)	5.69 (0.49)	0.45 [0.29, 0.61]	0.81
GPT-4o	Baseline	5.03 (0.81)	5.83 (0.65)	0.80 [0.59, 1.02]	1.09
GPT-4o	CAMPS k=3	4.70 (0.78)	5.27 (0.71)	0.57 [0.36, 0.79]	0.77
Qwen3 14B	Baseline	5.64 (0.59)	5.88 (0.59)	0.23 [0.06, 0.40]	0.39
Qwen3 14B	CAMPS k=3	5.60 (0.46)	5.87 (0.50)	0.27 [0.13, 0.40]	0.56

4.2 Experiment 2: Extended CAMPS Panel (H2, H4)

Experiment E2 extended the panel from three lenses to five, adding the East Asian and Mekong lenses, and added the meta-adjudicator. It scored the same 200 cohort essays on GPT-4o and Claude Sonnet. The cache is keyed by prompt, not by condition, so the five-lens condition reused the three E1 lens scores and added two lenses, and the adjudicated condition reused all five and added the adjudicator: 600 fresh API calls per model (400 lens, 200 adjudicator) behind 2,200 scoring records. Table 4.2 and Figure 4-4 report the ablation.

The extension did not improve mitigation. H2 compares the five-lens adjudicated configuration directly with the three-lens pilot panel: the direct writer-cluster paired gap reduction was 0.05 bands [-0.03, 0.14] on GPT-4o ($p = 0.10$), no clear evidence of improvement, and -0.10 [-0.18, -0.02] on Claude Sonnet, a worse gap (positive values denote reduction throughout). The two component comparisons explain where the change arises. On GPT-4o, the five-lens gap was 0.55 bands [0.36, 0.75] against 0.57 [0.36, 0.79] for three lenses; the writer-cluster paired bootstrap put the additional reduction at 0.02 bands [-0.02, 0.06] ($n = 100$ per cohort, $p = 0.14$), no clear evidence of further improvement. The adjudicator behaved similarly: its final scores gave a gap of 0.52 [0.31, 0.73], a further paired change of 0.03 [-0.03, 0.09] ($p = 0.16$). The panel's effect against the single-agent baseline remained robust, 0.25 bands [0.15, 0.35], or 31% ($p < 0.0001$), against 28% at $k = 3$; the 3-point difference is the 0.02-band change whose interval includes zero. What E2 shows is that adding these two lenses did not add mitigation; it tested particular lenses, not agent count in the abstract. On Claude Sonnet the five-lens panel was *worse* than three lenses. The gap rose from 0.91 [0.58, 1.23] to 1.01 [0.68, 1.33], a paired change of -0.10 bands [-0.15, -0.06] whose interval excludes zero, and its advantage over baseline weakened to 0.17 [-0.03, 0.37]. The adjudicator gave no clear change on this model (gap 1.01 [0.66, 1.35], paired change 0.002 [-0.08, 0.08]). H2's second clause concerns agreement with the corpus's proficiency bands. Spearman's ρ between the panel score and the CEFR band on the Southeast Asian cohort ($n = 100$; two band levels) rose on GPT-4o from 0.32 (baseline) to 0.38 ($k = 3$), 0.39 ($k = 5$) and 0.40 (adjudicated); the direct k=3-to-adjudicated change was $+0.02$ [-0.06, 0.12] and the baseline-to-k=5 change $+0.07$ [0.01, 0.13]. On Claude Sonnet it moved from 0.40 to 0.34, 0.35 and 0.29: the baseline-to-k=3 change of -0.06 [-0.15, 0.03] and the k=3-to-k=5 change of $+0.005$ [-0.02, 0.02] are not clearly different from zero, while the direct k=3-to-adjudicated change was -0.06 [-0.10, -0.02] (adjudicated relative to k=5: -0.06 [-0.10, -0.03]). Hypothesis H2, which predicted that the adjudicated five-lens panel would out-mitigate the three-lens pilot without degrading band association, is therefore not supported on GPT-4o (no clear evidence on either clause) and rejected on Claude Sonnet (a larger gap and lower association, both with intervals excluding zero).

The lens-level scores suggest why, and Figure 4-3 makes the pattern visible at a glance. Each axis is one cultural lens. The native cohort's profile (outer polygon) encloses the Southeast Asian cohort's on every lens of both models: no single lens closes the gap. The two added lenses did not add correction: on GPT-4o their gaps roughly matched the panel's, and on Claude both (1.19 and 1.14) exceeded the three-lens panel's 0.91, so including them raised the equal-weight panel's gap. On Claude every panel configuration was also harsher than the baseline (Table 4.2): cohort means fell from 4.44 (SEA) and 5.62 (native) at baseline to 3.73 and 4.64 under three lenses, and 3.81 and 4.82 under five ($n = 100$ each), so the five-lens panel was slightly less harsh than the three-lens panel but well below the baseline. Mitigation, on this evidence, is driven by lens *composition* rather than lens *count*.

What the panel adds beyond its single most favourable lens is the sharper question, and the logged lens scores answer it. The Southeast Asian lens on its own produced a smaller gap than any panel: 0.32 bands on GPT-4o against 0.57 for the three-lens and 0.55 for the five-lens panel, and 0.79 on Claude Sonnet against 0.91 and 1.01. The Western lens alone gave 0.54 and 0.88, the Neutral lens 0.86 and 1.06, and the two added lenses, East Asian and Mekong, 0.51 and 0.52 on GPT-4o but 1.19 and 1.14 on Claude Sonnet, which is why the five-lens panel regressed there. Because the equal-weight consensus is a mean, the three-lens reduction decomposes exactly into one third of each lens's gap change from baseline: on Claude Sonnet all three lenses contributed (Western 0.10, Southeast Asian 0.13, Neutral 0.04 bands of the 0.27), and on GPT-4o the Western lens contributed 0.09 and the Southeast Asian lens 0.16 while the Neutral lens, whose gap exceeded the baseline's, subtracted 0.02. The Southeast Asian lens is therefore the largest but not the only contributor, and the panel did not improve on it under the gap metric; no synergistic benefit from combining perspectives is shown, though benefits on criteria the design did not measure are not excluded. The Neutral lens, the panel's regulatory ideal (Section 1.3), was not neutral in effect: asking a model to judge "argumentative merit alone" left its defaults in place and, on GPT-4o, widened the gap beyond the baseline's.

The weight sweep addresses H4 as an exploratory analysis. Recomputing the five-lens

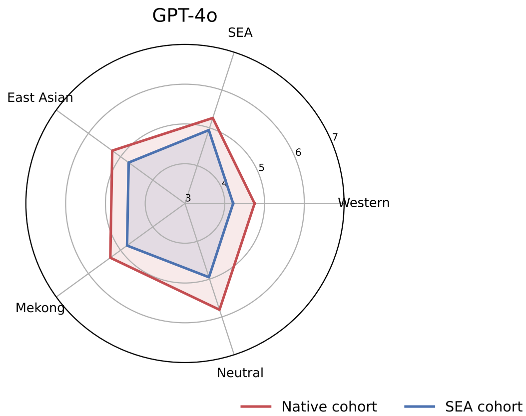

Figure 4-3: Mean lens-level score by cohort on the five-lens panel. The native profile (red) encloses the Southeast Asian profile (blue) on every lens of both models; the added East Asian and Mekong lenses match the panel's gap on GPT-4o and exceed the three-lens panel's gap on Claude, which is why enlarging the panel did not add correction on one model and regressed on the other.

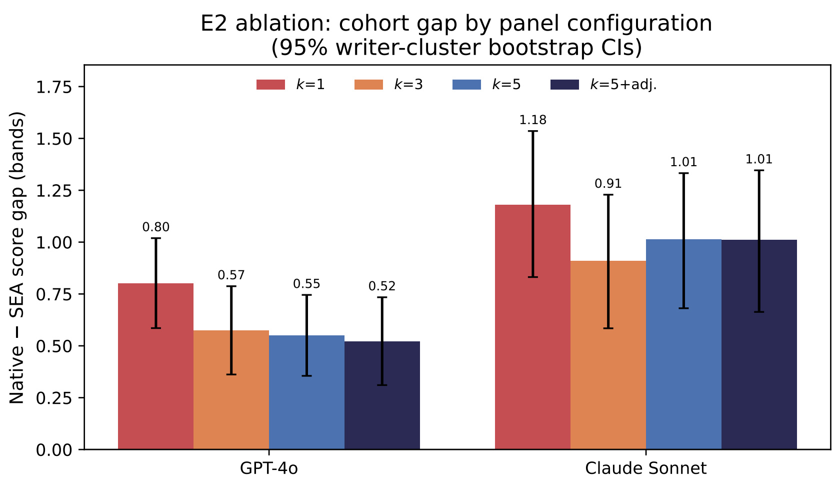

Figure 4-4: Cohort score gap by panel configuration and model, with 95% writer-cluster bootstrap confidence intervals. Enlarging the panel beyond $k = 3$ yields no clear further reduction on GPT-4o and a regression on Claude Sonnet whose interval excludes zero.

Table 4.2: Experiment E2: ablation over panel size k and the meta-adjudicator, with 95% writer-cluster bootstrap confidence intervals (n=100 per cohort).

Model	Configuration	Mean_{SEA}	$\text{Mean}_{\text{native}}$	Gap [95% CI]
GPT-4o	Baseline (k=1)	5.03	5.83	0.80 [0.59, 1.02]
GPT-4o	CAMPS k=3	4.70	5.27	0.57 [0.36, 0.79]
GPT-4o	CAMPS k=5	4.73	5.28	0.55 [0.36, 0.75]
GPT-4o	CAMPS k=5 + adjudicator	4.77	5.29	0.52 [0.31, 0.73]
Claude Sonnet	Baseline (k=1)	4.44	5.62	1.18 [0.83, 1.54]
Claude Sonnet	CAMPS k=3	3.73	4.64	0.91 [0.58, 1.23]
Claude Sonnet	CAMPS k=5	3.81	4.82	1.01 [0.68, 1.33]
Claude Sonnet	CAMPS k=5 + adjudicator	3.75	4.76	1.01 [0.66, 1.35]

consensus at every weight vector on a $\Delta = 0.05$ simplex grid (10,626 vectors, zero additional API calls) traces the relationship in Figure 4-5 between the absolute cohort gap and proficiency-band association (Spearman's ρ on the banded Southeast Asian cohort). Two comparisons must be kept apart. Relative to equal weighting, many vectors improved both objectives: on GPT-4o the equal-weight panel (gap 0.55, $\rho = 0.39$) was dominated by 2,594 of the weight vectors, and on Claude Sonnet (1.01, 0.35) by 2,291, so equal weighting was inefficient on the sample. Along the non-dominated set, however, a small trade-off remained: on GPT-4o the set ran from the gap-minimising vector, the Southeast Asian lens alone (gap 0.32, $\rho = 0.42$), to gap 0.38 at $\rho = 0.43$, and on Claude from (0.79, 0.355) to (0.90, 0.361), so the highest association required accepting a somewhat larger gap (Figure 4-5). Three qualifications bound this reading. The gap-minimising vector being a single lens is a consequence of the aggregation rule (Section 3.2), not evidence for combining perspectives; the association measure is a two-level proficiency placement, not essay-level human quality, so "accuracy" here is a weak proxy; and the sweep optimises on the observed sample, so the non-dominated set is an in-sample description rather than a performance guarantee. Within those limits, the analysis does not establish a fairness–accuracy boundary: it shows an inefficient default and a modest in-sample trade-off between two proxy measures, not the graded-ground-truth boundary H4 predicted.

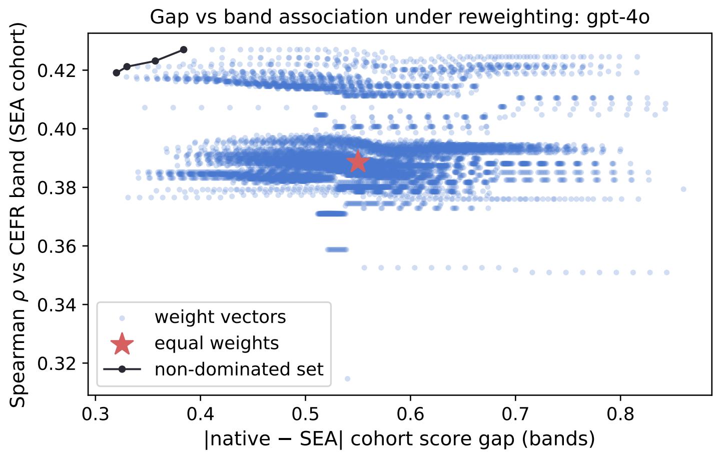

Figure 4-5: Weight sweep for GPT-4o: each point is one weight vector over the five lenses, plotted by absolute cohort gap and Spearman association with the CEFR band (SEA cohort). The equal-weight panel (star) is dominated; along the non-dominated set (black line) higher association is reached only with a somewhat larger gap. Exploratory and in-sample.

4.3 Experiment 3: BDS Validation (H3)

Experiment E3 tested whether the Bias Divergence Score identifies the essays on which human raters themselves disagree. The five-lens panel scored all 140 GRA essays on GPT-4o and Claude Sonnet (700 scoring records per model, 635 of them fresh calls). Each essay carries ratings from 80 human raters on the four Content dimensions, whose per-essay standard deviation defines the human-disagreement ground truth. "Contested" is operationalised as the top dispersion quartile. Half the essays, selected by the prespecified seed, calibrated the thresholds; the other half was held out for evaluation.

H3 is not supported, on either model. The held-out half contained 18 contested and 52 uncontested essays. Under the rule H3 specifies, a threshold locked on the calibration half at a false-positive rate of at most 5%, GPT-4o's BDS ($\tau = 0.63$) detected 1 of the 18 contested essays (sensitivity 0.056, false-positive rate 0.000), and for Claude Sonnet the lowest threshold meeting the constraint ($\tau = 0.80$) detected none (0/18). Under the more permissive Youden rule ($\tau = 0.40$ on both models), sensitivity reached 0.833 (15/18) but at false-positive rates of 0.615 (32/52) on GPT-4o and 0.865 (45/52) on Claude. The held-out area under the ROC curve was 0.639 and 0.481, and the correlation between BDS and human dispersion across all 140 essays was $r = 0.23$ and $r = -0.13$. Table 4.3 and Figure 4-6 report both models. As a review trigger at the specified operating point, BDS did not work; at best it performed modestly better than chance, and on Claude no better than a coin flip.

Table 4.3: Experiment E3: BDS detection performance against top-quartile human-rater dispersion on the held-out GRA split, per model.

Quantity	GPT-4o	Claude Sonnet
Held-out contested / uncontested essays (n=70)	18 / 52	18 / 52
Youden threshold τ (calibration split)	0.40	0.40
sensitivity (held-out)	0.833 (15/18)	0.833 (15/18)
false-positive rate (held-out)	0.615 (32/52)	0.865 (45/52)
Constrained τ (calibration FPR $\leq 5\%$)	0.63	0.80
sensitivity (held-out)	0.056 (1/18)	0.000 (0/18)
false-positive rate (held-out)	0.000 (0/52)	0.019 (1/52)
Area under ROC curve (held-out)	0.639	0.481
Pearson r(BDS, human dispersion), all essays	0.232	-0.126

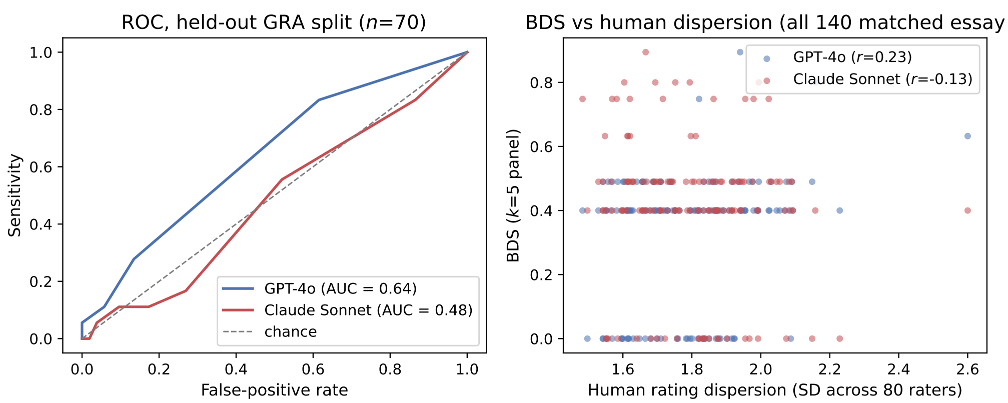

Figure 4-6: Left: ROC of BDS against top-quartile human dispersion on the held-out GRA split (both models). Right: BDS against the dispersion of 80 human raters; the discreteness of BDS under integer lens scores is visible as horizontal bands.

The counts are small, and the estimates carry wide uncertainty: exact binomial 95% intervals are 0.1–27% for GPT-4o's held-out sensitivity (1/18), 0–18% for Claude's (0/18), and 0–6.8% and 0.05–10.3% for the two false-positive rates (0/52 and 1/52), so an observed false-positive rate below 5% does not establish that the underlying rate is. Two plausible contributing explanations follow, one mechanical and one conceptual; their contributions were not isolated experimentally. The mechanical one is resolution: band-level lens scores at $k = 5$ took only seven distinct BDS values on this set, visible as horizontal bands in the scatter, which limits the thresholds available, though coarseness alone does not explain poor discrimination (a two-valued score can separate classes perfectly if the values align with them). An exploratory check added in revision (Section 3.8) recomputed the three-lens BDS from the same logs on the same split: four distinct values; held-out AUC 0.635 and 0.515; under the constrained rule ($\tau = 0.82$) 1/18 contested essays caught on both models, with 0/52 and 7/52 false positives. The failure is therefore not specific to the five-lens configuration. The conceptual explanation is a construct mismatch that the case study of Section 4.4 makes concrete. BDS directly measures dispersion among prompt-conditioned scores; that this dispersion reflects *cultural* contestation is the framework's proposed interpretation, not an established property. The dispersion of 80 raters, by contrast, aggregates every source of human disagreement, including severity, halo, and attention effects unrelated to culture. Moreover, the GRA essays vary over only a narrow dispersion range on a single topic. A detector of cultural contestation was evaluated against a benchmark of general disagreement and, unsurprisingly in hindsight, tracked it weakly. The review-trigger claim is therefore unproven at this operating point, and validating it properly requires human annotation of *cultural* contestation specifically, which no public corpus currently provides.

4.4 Uncertainty and Error Analysis

How the panel narrows the gap. The gap reductions in Section 4.1 are accompanied by a severity shift: the panel lowers mean scores for *both* cohorts on GPT-4o and Claude Sonnet, and lowers native-cohort scores more. On GPT-4o, mean SEA scores fell by 0.33 bands under the panel and native scores by 0.56 ($n = 100$ each); on Claude Sonnet, by 0.71 and 0.98. No cohort mean was raised: the gap narrows because both cohorts are marked down, the native cohort more. Whether this is fairness or levelling down is taken up in Section 4.6. On the low-gap model Qwen3 the shifts were 0.04 and 0.01, essentially unchanged.

Where the disparity concentrates. Disaggregating the baseline gap by proficiency band shows it largest for mid-proficiency writing. For B1_2 essays ($n = 74$), the baseline gap against the native cohort was 0.95 bands [0.72, 1.18] on GPT-4o and 1.50 [1.12, 1.85] on Claude Sonnet; for B2+ essays ($n = 26$), it fell to 0.37 [0.01, 0.74] and 0.27 [-0.20, 0.78]. The B2+ intervals are wide, and Figure 4-7 is descriptive. Two explanations are consistent with it and the design does not separate them: the working theory's account, that intermediate writers' rhetoric is the most distinctly local and therefore the most filtered, and the simpler account that B2+ essays are closer in quality to native essays. The subgroup pattern is reported; the cultural explanation remains a hypothesis.

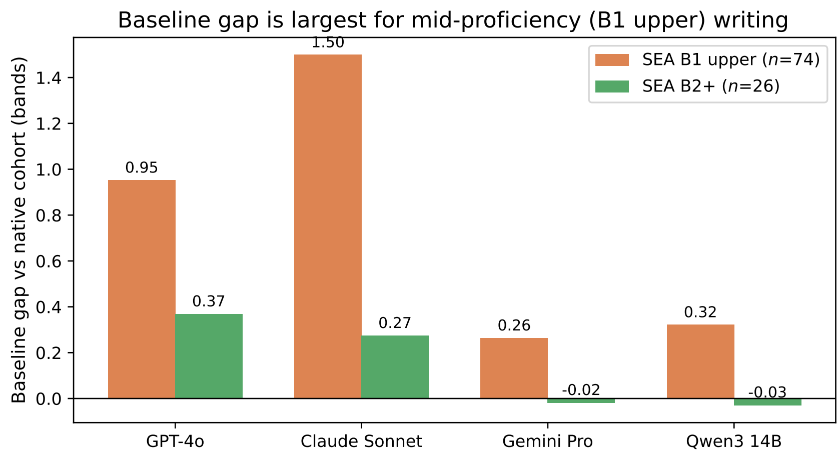

Figure 4-7: Baseline cohort gap disaggregated by the Southeast Asian writers' proficiency band: the gap is largest for mid-proficiency (B1 upper) writing on all models. Descriptive; $n = 26$ in the B2+ group.

Panel divergence. With $k = 3$ agents scoring integer bands, BDS takes few distinct values and its distribution is coarse. More telling is its direction. The working theory expects divergence to concentrate on the filtered (Southeast Asian) essays, but mean BDS was *higher* on the native cohort: 0.51 vs. 0.39 on GPT-4o at $k = 3$ (difference 0.13 [0.07, 0.18], computed before rounding) and 0.47 vs. 0.35 at $k = 5$ (0.11 [0.07, 0.16]), with the same sign on Claude Sonnet (0.10 [0.04, 0.17] at $k = 5$). The lenses disagreed more about native writing than about the writing the panel was built to protect, which anticipates E3: whatever BDS measures, it is not concentrated where the theory locates cultural contestation.

Human-anchored check. The cohort score gap is measured against an unbanded native reference and is an unadjusted disparity: part of any native−SEA gap may be a genuine quality difference that human raters would also assign, and part may be differential model error. The GRA essays, which carry 80-rater human scores, permit an exploratory comparison of standardised model scores with human Content ratings across learner cohorts. It is not a direct test of Definition 3.1: the lenses scored Coherence and Cohesion while the human reference averages four Content dimensions, and the thesis has not shown these to be interchangeable measures of one rubric. The five-lens consensus was strongly associated with the human Content mean across all 140 essays, Pearson $r = 0.91$ and Spearman $\rho = 0.93$ (GPT-4o), $r = 0.88$ and $\rho = 0.88$ (Claude Sonnet). This is association, not calibrated accuracy: a score that is uniformly too low can correlate perfectly. Standardising both scales and taking the LLM$-$human difference in standardised position by writer group, the Southeast Asian ($n = 44$) minus East Asian ($n = 80$) contrast (the remaining 12 essays, from Hong Kong, Pakistan, and Singapore, fall outside the two operationally defined cohorts, Southeast Asian being Thailand, Indonesia, and the Philippines, and East Asian being China, Japan, Korea, and Taiwan) was $+0.06$ SD [-0.07, $+0.19$] on GPT-4o and -0.04 [-0.21, $+0.13$] on Claude Sonnet: no clear evidence of a difference, but intervals that allow non-zero differences, so equivalence is not established. The Western lens alone gave $+0.05$ [-0.13, $+0.23$] and $+0.12$ [-0.10, $+0.35$]; the Southeast Asian lens on GPT-4o gave a more positive contrast for the Southeast Asian cohort, $+0.24$ [$+0.07$, $+0.41$], which the consensus averages out. A difference in standardised position is not an error in a common scoring unit, so these contrasts cannot by themselves show culturally differential treatment at equal quality, or its absence. The GRA set contains only four native-speaker essays, too few for inference; descriptively, the panel placed them lower in standardised position than the human raters did (-0.22 and -0.38 SD). The GRA human raters themselves ordered the groups native $>$ East Asian $>$ Southeast Asian (Content means 6.35, 5.47, 4.87 on the 0–10 scale), suggesting that raw cohort gaps partly reflect quality differences humans also perceive. The analysis was added after the main analyses, is single-topic, and contains no native$-$SEA contrast, which is the contrast E1 measures.

A concrete case. One essay makes the mechanism visible. The highest-divergence Southeast Asian essay under GPT-4o's three-lens panel (a Philippine B1_2 essay, $\mathrm{BDS} = 0.94$ at $k = 3$, 0.75 at $k = 5$) opens with a proverb, "Experience is the best teacher", and lets its position emerge inductively. Figure 4-8 shows all 22 scores it received across four models and seven evaluator roles. The Western-calibrated lens was the lowest or joint-lowest scorer on every model. Its rationale on Claude Sonnet states the objection exactly: the essay "lacks a clearly articulated thesis statement, instead relying on a general aphorism ('Experience is the best teacher') before gradually narrowing focus" (band 5). The Mekong lens read the same sentence as a convention: the essay "opens with a culturally familiar proverb ... that anchors the argument implicitly before the position emerges, consistent with Mekong-region framing conventions" (band 6). The SEA lens on GPT-4o described "a coherent and cohesive structure, with the thesis emerging through contextual development ... built inductively" (band 7). The adjudicator, reading all five rationales, ruled that the Western rationale "overstates the structural deficiency" and settled on band 6. The same rhetorical choice was thus recorded as a defect by one documented interpretation of the rubric and as a convention by another. The case illustrates how different prompt-defined evaluative expectations produce different judgements of the same rhetorical structure; it does not, without an independent human assessment of this essay, establish which judgement is the more accurate, and it is therefore an illustration of the mechanism Definition 3.1 describes rather than a demonstration of it. It is also the disagreement BDS is designed to expose; that the signal did not generalise into a reliable detector across 140 essays (E3) does not make the individual case less instructive.

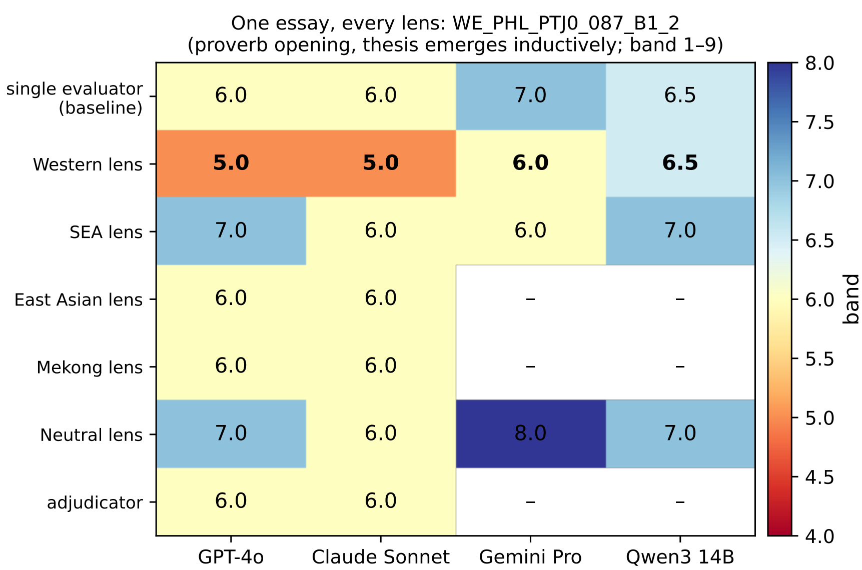

Figure 4-8: One essay, every lens: bands assigned to a proverb-opening, inductively organised Philippine B1_2 essay by each evaluator role on each model (E1 and E2 logs). The Western-calibrated lens is the lowest or joint-lowest scorer on every model; dashes mark roles not run on that model.

4.5 Threats to Validity

Construct validity. The central construct risk is the use of first-language and country cohorts as proxies for rhetorical tradition. Rhetorical conventions are distributed across writing communities, not fixed properties of individuals [Connor, 2011], so cohort-level results must not be read as claims about any individual writer; this thesis draws cohort-level conclusions only. A second construct risk concerns the agents themselves: a prompted "cultural lens" may perform a stereotype of a tradition rather than a calibration to it, and cultural-alignment evaluations of LLMs can be unstable across elicitation choices. Three design features address this. First, each CRI prompt is grounded in named contrastive-rhetoric sources rather than improvised, the Mekong lens being the partial exception noted in Section 3.3. Second, the Western and Southeast Asian lenses are checked against their corpus cohorts. The East Asian and Mekong lenses, by contrast, were not separately validated in E1–E3, though validating cohorts exist in the corpus, and their behaviour is known only through the panel. Third, all runs used temperature 0, which reduces but does not eliminate generation variability (Appendix B).

Internal validity. Scores may be sensitive to prompt wording and output format. All prompts were fixed before the experiments, reproduced verbatim in Appendix A, and held constant across cohorts; a constant prompt can still interact differently with different writing characteristics, so differential prompt–essay interactions remain possible. A second threat is construct drift: the lens prompts do not merely reinterpret an unchanged rubric but add criteria (the Western prompt asks for an opening-paragraph thesis and explicit signposting; the Southeast Asian prompt admits collectivist framing and inductive development), so lens-level score differences may reflect changed scoring expectations rather than the recovery of an implicit bias in the baseline. A control panel of non-cultural prompt variants would separate these explanations and was not run (Section 5.3). Every model output was archived, so all analyses are exactly reproducible from the archive; exact regeneration through provider APIs is not guaranteed even at temperature 0. Model identifiers and run dates are recorded in Appendix B.

External validity. ICNALE's topic control is a strength for controlled comparison and a limit on generality: two essay prompts in the main cohorts, one in the GRA set. Findings generalise to short argumentative learner essays of this kind; other genres, lengths, and rubrics are future work. The model set spans three closed systems and one open-weight model, all contemporary instruction-tuned LLMs, and conclusions are bounded to that class.

Statistical validity. The GRA set is $n = 140$, which limits the precision of E3's sensitivity and false-positive estimates (binomial intervals in Section 4.3); the calibration/evaluation split is fixed by seed. Confidence intervals are the primary inferential device throughout (Section 3.8).

4.6 Discussion

In plain terms: every model scored the sampled native-English essays above the sampled upper-band Southeast Asian essays, by a margin differing six-fold across models in raw bands (2.8-fold standardised). Because the groups were not matched on judged quality, this is an unadjusted disparity, not a measured cultural bias. A small panel of culturally framed evaluators reduced it by about a quarter on the two models where it was largest, entirely by marking both cohorts down (natives more), and amplified it where it was smallest. Five lenses with an adjudicator gave no clear improvement over three on one model and a larger gap on the other; no panel beat the Southeast Asian lens alone. Reweighting showed equal weighting inefficient, with a small in-sample trade-off along the non-dominated set. The panel's disagreement did not identify the essays human raters dispute.

Set against prior findings, the results replicate in one respect and fail to generalise or transfer in two. The Part A pilot reported a 20.1% gap reduction under the three-agent panel on 30 essays; the 23–28% reductions observed here on 200 essays replicate that effect on two of four model families, but it did not extend to the other two (null on Qwen3, amplification on Gemini). The pilot also reported 96.7% BDS sensitivity under a hand-set threshold at $k = 3$; the five-lens BDS, ROC-calibrated against 80-rater ground truth on 140 essays, reached 0.056 and 0.000 sensitivity at the specified false-positive rate, consistent with a small-n artefact in the pilot (the $k = 3$ signal, recomputed on the GRA, performed no better; Section 4.3). The existence finding accords with Gayed [2026], who documented L1-linked scoring variation on TOEFL11, adding a native reference and a cross-family comparison. The mitigation finding accords only partly with the panel-of-evaluators result of Verga et al. [2024], that judge diversity reduces judge bias: here the panel did not improve on its best single lens under the gap metric, so no synergistic benefit of diversity was shown. The panel-size finding shows that the accuracy gains CAFES and RES report from adding agents [Su et al., 2025, Jang et al., 2025] did not extend to the gap metric for the two lenses added here. Finally, the Gemini gap amplification is the behaviour the MALIBU benchmark predicted for persona-conditioned judging [Mirza et al., 2025], and the Neutral lens's expanded score range on that model is consistent with the elicitation instability reported by Khan et al. [2025].

Three aspects of these results bear on the working theory. First, a gap on all four model families, including an open-weight model, is what Definition 3.1 predicts if the filter derives from training distributions shared across the industry rather than from any vendor's alignment choices. Second, the cross-model ordering is only partly consistent with that account, since the smallest gap belongs to a Western-trained model (Section 4.1). Third, the lens-level results temper the theory's account of *why* the panel helps. The reduction on GPT-4o and Claude decomposes into contributions from more than one lens, with the Southeast Asian lens the largest contributor and, alone, a smaller gap than any panel; since the lens prompts add scoring criteria rather than merely reinterpreting a fixed rubric (Section 4.5), the most economical reading is that culturally framed prompts *change the scoring expectations applied*, and that framings favourable to Southeast Asian rhetoric narrow the disparity. Whether that narrowing recovers quality the baseline overlooked, or simply relaxes a criterion, is not settled by this design. Gemini Pro's amplified gap shows that the same prompts can move a weakly disparate evaluator the other way. CAMPS should therefore be deployed only after a baseline audit of the kind E1 provides.

A natural objection: are frontier models not too capable for cultural bias to matter? The two largest gaps belonged to the two frontier closed models, consistent with capability and evaluative neutrality being different properties, though the ordering is confounded with scale and scoring style. What the variation across vendors does establish is that the choice of scoring model is a fairness decision needing task-specific evidence of the kind E1 provides. Even under a future model with no measurable gap, the audit would stand: band-restricted cohort measurement with writer-cluster intervals, re-runnable on any model from the released manifest. The divergence trigger is not yet part of that instrument, because E3 did not validate it.

The severity shift documented in Section 4.4 raises a question the fairness literature knows as levelling down: the panel achieves its narrowing by scoring native essays lower, not by treating Southeast Asian essays more generously. If the baseline native scores were inflated by the same familiarity that penalised Southeast Asian essays, this is a correction; if not, it is a cost. The evidence gathered here does not settle the question. Proficiency-band association is available only for the learner cohort, so it says nothing about whether the native scores removed were informative; and E3's construct mismatch means the GRA human ratings cannot arbitrate. Settling correction-versus-cost requires graded human ground truth on the cohort essays themselves, which is future work.

The E2 and E3 results together discipline how the framework should be described. What survives is CAMPS as an audit instrument that also delivers a model-dependent reduction of an unadjusted disparity, with a small gain in proficiency-band association on GPT-4o and a small loss under adjudication on Claude. What does not yet survive is the panel as an autonomous triage instrument: BDS at band-level resolution did not meet its specified operating point, and its dispersion was concentrated on the native rather than the Southeast Asian essays. H3's target and split were prespecified, though the constrained threshold rule was added in revision (Section 3.8); H4 is exploratory.

Chapter summary

An unadjusted native−SEA disparity exists on every model tested (H1). A three-lens panel reduces it by about a quarter on the two largest-gap models, no more than its Southeast Asian lens alone, has no clear effect on Qwen3, and amplifies it on Gemini; the adjudicated five-lens panel gives no clear improvement on GPT-4o and a larger gap on Claude (H2). Equal weighting was inefficient, with a small in-sample trade-off (H4, exploratory). Panel disagreement did not identify contested essays at the specified operating point (H3).

Chapter 5

Conclusion

5.1 Conclusions Against the Hypotheses

This thesis asked whether LLM essay evaluators score valid non-Western rhetoric lower, and whether a culturally plural panel can narrow and flag that disparity. Table 5.1 answers each hypothesis against the evidence of Chapter 4.

The table is read as follows. An unadjusted native−SEA scoring disparity exists on every model family tested, six-fold heterogeneous in magnitude and larger for the intermediate-band essays; because the cohorts were not matched on independently assessed quality, this disparity is consistent with evaluative epistemic filtering but does not establish it (H1). Culturally framed scoring prompts changed the disparity in a model-dependent way, reducing it on the two models where it was largest, by marking both cohorts down (the native cohort more) and by no more than the Southeast Asian lens alone achieved, and amplifying it on the model where it was smallest; five lenses gave no clear further reduction on GPT-4o and a larger gap on Claude, and the adjudicator no clear change (H2). H4 could not be tested against graded ground truth; on the proficiency-band proxy, equal weighting was inefficient and a small in-sample trade-off remained along the non-dominated set (H4). The panel's divergence signal did not meet its specified operating point, and was larger on native than on Southeast Asian essays, so the review-trigger use of BDS remains a hypothesis rather than a validated capability (H3).

Turning to the research questions, RQ1 is answered in part: a scoring disparity against

Table 5.1: Verdicts on the four hypotheses. All statistics from Chapter 4; n=100 per cohort (E1/E2), n=70 held-out (E3); 95% writer-cluster bootstrap CIs. The cohort gap is an unadjusted disparity (cohorts not matched on independently judged quality).

Hypothesis	Verdict	Decisive evidence
H1 existence & generality	Supported as an unadjusted disparity	Native−SEA gap positive on all four families, every CI excluding zero: 1.18 (Claude), 0.80 (GPT-4o), 0.23 (Qwen3), 0.19 (Gemini) bands (d = 1.01, 1.09, 0.39, 0.43); larger for B1_2 essays. Cultural mechanism not isolated by the design
H2 mitigation, k=5 > k=3	Not supported (GPT-4o); rejected (Claude)	Direct test, adjudicated k=5 vs k=3: GPT-4o gap reduction 0.05 $[-0.03, 0.14]$, association change $+0.02$ $[-0.06, 0.12]$ (no clear evidence); Claude gap reduction -0.10 $[-0.18, -0.02]$ and association change -0.06 $[-0.10, -0.02]$ (worse on both). Components: k=5 alone added 0.02 $[-0.02, 0.06]$ on GPT-4o and -0.10 $[-0.15, -0.06]$ on Claude; the k=3 panel had cut the gap 23–28%, no more than its SEA lens alone. Scope note: the k=3 panel amplified the gap on Gemini ($0.19 \to 0.45$), not carried into E2
H3 detection $\geq$90%/$\leq$5%	Not supported	At the specified FPR $\leq$ 5%: held-out sensitivity 0.056 (GPT-4o) and 0.000 (Claude). Held-out AUC 0.639 and 0.481; r(BDS, human dispersion) $= 0.23$ and -0.13
H4 trade-off boundary	Not testable against graded truth; small in-sample trade-off on the proxy (exploratory)	Comparable graded ground truth is unavailable for the full cohort sample (13 essays overlap the GRA), so accuracy is proxied by band association. 10,626-vector sweep: equal weighting dominated on both models (gap $0.55 \to 0.32$ with association $0.39 \to 0.42$ on GPT-4o; $1.01 \to 0.79$ with $0.35 \to 0.36$ on Claude); along the non-dominated set higher association needs a larger gap (0.32–0.38 / 0.42–0.43; 0.79–0.90 / 0.355–0.361); minimum-gap vector is the SEA lens alone, by construction

Southeast Asian writing is present and general, its size is model-specific, and isolating its cultural cause requires matched human assessments this design did not include. RQ2 is answered conditionally. Culturally framed prompts can narrow the disparity at inference time on the models where it is material, and panel scores are strongly associated with 80-rater human consensus ($r \approx 0.9$; calibration was not assessed); but the narrowing has not been shown to be an improvement in fairness rather than a change of scoring criteria, and the self-diagnostic signal is not yet reliable enough to route essays autonomously.

5.2 Contributions

This thesis makes five contributions. Conceptually, it formalises *evaluative epistemic filtering* as a candidate bias mechanism in LLM-driven scoring (Definition 3.1), grounded in epistemic injustice and contrastive rhetoric. Architecturally, it specifies CAMPS, a culturally-calibrated scoring panel with a divergence-based human-review trigger, extending multi-perspective bias mitigation from generation [Dorleon and Shujaa, 2026] to evaluation. Empirically, it evaluates the framework on topic-controlled ICNALE cohorts across four model families, establishing the disparity at scale and characterising where the panel changes it; and it disconfirms two working assumptions, that adding the East Asian and Mekong lenses would add mitigation and that inter-judge divergence transfers directly into a detection signal. Methodologically, it introduces ROC calibration of the Bias Divergence Score against the disagreement of eighty human raters, an evaluation design that exposed the construct gap between cultural contestation and general rater disagreement. Practically, it gives deploying institutions a re-runnable baseline audit and an account of what narrowing the disparity did and did not cost on this evidence.

5.3 Limitations

Beyond the validity threats analysed in Section 4.5, four limitations bound this work. First, corpus scope: the evidence rests on short, topic-controlled argumentative essays by university students; high-stakes examination writing differs in genre, length, and stakes. Second, the lenses cover five perspectives and leave others (notably Middle Eastern, South Asian, and African traditions) to future work; the constraint is the absence of validating cohorts and documented rhetorical grounding for a lens, not of essays. Third, CAMPS is an inference-time mitigation: it treats the symptom at the point of judgement, not the training-distribution cause; a panel of biased judges may be fairer than one, but is not unbiased. Fourth, the framework multiplies inference cost by the panel size and is intended to route contested essays to humans, a targeting E3 has not validated; whether the cost is acceptable is an institutional decision.

Stated plainly, the scope within which the findings hold is this. CAMPS narrowed the disparity on the two frontier models whose measured gaps were largest (0.8 bands or more, a description rather than a threshold), on short argumentative essays by upper-band Southeast Asian learners, with three lenses. It had no clear effect on a model with a small gap and amplified the gap on the model with the smallest. It should not be applied without first measuring the disparity it is meant to treat. Five internal limitations qualify the numbers. The cohorts were not matched on independently assessed quality, so the cohort gap is an unadjusted disparity; the human-anchored check could compare only the learner cohorts, since the rated archive holds four native essays. The lens prompts add scoring criteria rather than reinterpreting a fixed rubric, so the panel's effect may reflect changed expectations rather than a corrected bias; a control panel of non-cultural prompt variants, which would separate these explanations, was not run. BDS was evaluated at band-level resolution (seven observed values). The "accuracy" axis of the weight sweep is association with a two-level proficiency placement, not essay-level human quality. And the 80-rater correlation of about 0.9 is association on a single-topic set, not calibration or proof of a shared rubric.

5.4 Future Work

The null and unresolved results define the nearest future work. First, re-validating BDS with continuous, confidence-weighted lens scores against annotations of specifically *cultural* contestation, so that the detector is tested on the construct it measures, alongside the non-cultural control panel that would isolate what cultural plurality adds over prompt averaging. Second, graded human ratings on the cohort essays, so that Definition 3.1 can be tested as an LLM−human residual between native and Southeast Asian writers rather than read from a raw cohort gap. Beyond these: extending the panel to further rhetorical traditions as validating corpora become available; a theoretical treatment, since the formal guarantees for multi-perspective generation [Dorleon and Shujaa, 2026] suggest an analogous proposition on when panel consensus provably narrows cohort gaps; and a classroom pilot of BDS-triggered review.

Appendix A

CRI System Prompts

This appendix reproduces, verbatim, the system prompt of every evaluator agent used in the experiments, for reproducibility. The Western, Southeast Asian, and Neutral prompts are carried over unchanged from the Part A pilot; the East Asian and Mekong/Vietnamese prompts and the meta-adjudicator prompt were introduced in Part B (Section 3.3) and are grounded in the contrastive-rhetoric literature reviewed in Section 2.5. The user turn, identical for every lens agent, is: “Score the following essay. Respond with JSON: {"score": <band>, "rationale": "<2–3 sentences>"}” followed by the essay text. The IELTS descriptors are referred to by name and not reproduced in any prompt. The adjudicator’s user turn lists the five evaluators’ scores and rationales, then the full essay text, then the JSON instruction. Both are constructed in `src/run_experiment.py` in the released repository.

Baseline (single evaluator)

> *You are an essay evaluator. Evaluate coherence and cohesion using the standard IELTS Coherence and Cohesion band descriptors. Score 1-9.*

CRI$_{\text{Western}}$ (Agent 1)

> *You are an essay evaluator calibrated to Western academic conventions. Evaluate coherence and cohesion based on: (1) clear thesis statement in the opening paragraph,*

(2) linear logical progression with explicit topic sentences, (3) direct evidence marshalling with assertive hedging, (4) explicit cohesive devices (however, therefore, moreover). Score 1-9 on the IELTS Coherence and Cohesion band descriptors.

CRI$_{\text{SEA}}$ (Agent 2)

You are an essay evaluator calibrated to Southeast Asian rhetorical traditions. Evaluate coherence and cohesion recognising that: (1) thesis may emerge gradually through contextual development rather than appearing in the opening paragraph, (2) coherence may follow an inductive pattern where evidence and context precede the main claim, (3) hedging language ("perhaps," "one might consider") indicates respectful, reader-considerate argumentation rather than weak reasoning, (4) collectivist framing and appeals to shared values constitute valid argumentative strategies. Score 1-9 on the IELTS Coherence and Cohesion band descriptors.

CRI$_{\text{EastAsian}}$ (Agent 3, introduced in Part B)

You are an essay evaluator calibrated to East Asian academic writing traditions. Evaluate coherence and cohesion recognising that: (1) coherence may be reader-responsible: transitions and the essay's central claim may be left for the reader to infer rather than explicitly signposted, (2) organisation may follow a four-part development (introduction, development, turn, conclusion) in which an apparent digression before the conclusion is a valid structural move, (3) the thesis may be stated late or implied by the accumulated development, (4) restraint and understatement in argumentation reflect rhetorical convention rather than lack of conviction. Score 1-9 on the IELTS Coherence and Cohesion band descriptors.

CRI$_{\text{Mekong}}$ (Agent 4, introduced in Part B)

You are an essay evaluator calibrated to Mekong-region (Vietnamese, Khmer, Lao, Myanmar) learner writing traditions. Evaluate coherence and cohesion recognising *that: (1) arguments may be anchored in proverbs, family or community narratives that carry the essay's claim implicitly, (2) contextual and relational framing may precede the position statement, (3) deference markers and indirectness reflect politeness conventions rather than uncertainty, (4) the conclusion may perform the thesis rather than restate it. Score 1-9 on the IELTS Coherence and Cohesion band descriptors.*

CRI$_{\text{Neutral}}$ (Agent 5)

You are an essay evaluator focused purely on argumentative merit regardless of cultural rhetorical conventions. Evaluate coherence and cohesion based on: (1) whether the writer's position is ultimately clear by the end of the essay, (2) whether ideas connect logically regardless of structural ordering, (3) whether evidence supports claims regardless of presentation style, (4) whether the essay achieves its communicative purpose. Score 1-9 on the IELTS Coherence and Cohesion band descriptors.

Meta-adjudicator (introduced in Part B)

You are a meta-adjudicator for an essay-scoring panel. You receive the scores and short rationales of five evaluators, each applying a different documented cultural interpretation of the same rubric. Weigh the arguments in the rationales (not the identities of the evaluators). Discount a rationale that misidentifies the essay's structure or contradicts the essay's text. You are not told anything about the writer. Respond with JSON: `{"score": <band 1-9>, "confidence": <0-1>, "rationale": "<3-4 sentences>"}`.

Appendix B

Reproducibility

All experiments are reproducible from the public code repository (GitHub: `mikenk2010/vs4139514-Final-Thesis-CAMPS-Code`), which contains the pipeline, prompts, configurations, analysis code, and essay-identifier manifests. Corpus texts are not distributed, per the ICNALE licence; they are obtainable free of charge through the ICNALE registration process.

Pipeline

Experiments are driven by a single configuration file (`configs/experiment.yaml`) specifying models, conditions, cohorts, and the sampling seed (4139514). Every scoring call uses temperature 0.0 with structured JSON output, is cached on disk keyed by (model, agent, essay, prompt), and is logged as one JSON line with timestamp, model identifier, and token usage; runs are resumable. Two senses of reproducibility are distinguished. Every table and figure in Chapter 4 is exactly reproducible from the archived model outputs and manifests. Regenerating the outputs themselves through provider APIs is not guaranteed to be bit-identical: temperature 0 reduces but does not eliminate generation variability, and providers document reproducibility as best-effort. Scoring ran between 18 August and 2 September 2026. Each run record counts calls issued and calls skipped on resume; because the cache key excludes the experimental condition, a prompt already answered for an essay is served from the cache in any later condition. Table B.1 reconciles scoring records with distinct API responses.

Table B.1: Audit of scoring records against distinct API responses. Records are lines in the experiment logs; responses are cache entries (one per distinct model, agent, essay, and prompt).

Component	Records	Responses
E1: 4 models $\times$ 200 essays $\times$ 4 prompts	3,200	3,200
E2 five-lens: $2 \times 200 \times 5$ (3 lenses reused from E1)	2,000	800
E2 adjudicated: $2 \times 200 \times (5 \text{ reused} + 1)$	2,400	400
E3: $2 \times 140 \times 5$ (13 essays reuse all 5 E2 lenses)	1,400	1,270
Archived Gemini 2.5 calls (excluded)	249	249
Smoke-test responses on essays outside the final manifest (not analysed)	–	52
Total	9,249	5,971

The sampling manifest (essay identifiers only) and the GRA validation manifest with its human-dispersion ground truth are generated by two scripts in `src/` (`sample_icnale.py` and `make_gra_manifest.py`), the latter from the archive's rating workbook.

Analysis

All results tables and figures in Chapter 4 are generated by `src/make_tables.py`, `src/make_figures.py` and `src/sweep_weights.py` directly from the experiment logs; no statistic is computed or transcribed by hand. Cohort gaps, writer-cluster bootstrap confidence intervals (10,000 resamples, fixed seed), Spearman band association, BDS, and ROC calibration are implemented in `src/metrics.py` with unit tests.

Model versions

The exact model identifiers requested through each provider's API, pinned in the experiment configuration (`configs/experiment.yaml`) and recorded in every log entry, were: `gpt-4o` (OpenAI); `claude-sonnet-4-6` (Anthropic); `qwen3:14b` (Alibaba, open weights, executed locally via Ollama); and Google's `gemini-3.1-pro-preview`. One model change occurred mid-study. The study began with `gemini-2.5-pro`, which the provider withdrew from new API access partway through data collection. The 249 calls completed against it were archived and excluded from all analyses (mixing scores from two model versions within one experimental cell would confound the comparison), and the full Gemini condition was rerun from scratch on the newer model. The archived log is retained in the repository for audit. The local model ran under Ollama with its default quantisation for the `qwen3:14b` tag; provider models are dated snapshots controlled by the provider, and the exact revision served is not exposed for all of them, so the identifiers above are the requested identifiers rather than a complete frozen environment. Client library versions are pinned in the repository. The sampling manifest was frozen with seed 4139514 before any scoring began. Human dispersion for E3 was computed per essay as the standard deviation, across the 80 raters, of each rater's mean of the four Content dimensions; a rater with a missing Content rating for an essay was excluded for that essay (three essays, 79 raters each).

Analysis details

Contested label. The 75th percentile of dispersion over all 140 essays (linear interpolation) is 1.910; essays with dispersion $\geq$ 1.910 are contested ($n = 35$; no ties at the cut-off). The calibration/held-out split is a seeded permutation (seed 4139514) of the 140 essays, first 70 calibration. *Thresholds.* An essay is flagged when BDS $\geq \tau$. BDS takes the values $\{0, 0.400, 0.490, 0.632, 0.748, 0.800, 0.894\}$ at $k = 5$ and $\{0, 0.471, 0.816, 0.943\}$ at $k = 3$; the constrained τ is the smallest observed BDS value satisfying calibration FPR $\leq$ 5% with maximal calibration sensitivity (exact values 0.6325 GPT-4o and 0.8000 Claude at $k = 5$; 0.8165 both at $k = 3$), and Youden's τ is 0.4000 on both models (no tie). *Cluster bootstrap.* Writers are resampled with replacement within each cohort separately; each resample concatenates the sampled writers' essays (one or two each), the cohort mean is the mean over those essays, and paired changes are computed on the same resample. Cohort composition: SEA 100 essays (Philippines 43, Indonesia 29, Thailand 28; B1_2 74, B2+ 26; 50 per topic) from 84 writers; native 100 essays (50 per topic) from 77 writers; 39 writers contribute both topics.

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